

uMap-Logger

Version 1.08.003 build006



ORIGINAL INSTRUCTION IN ENGLISH LANGUAGE



LEGAL NOTICE

Use of the Manual

This Manual is intended to be used solely by authorized user of IDS GeoRadar Systems. It is specific for internal business purpose only. If you are not an authorized user, you are NOT allowed to read this Manual and you must return it to IDS GeoRadar as soon as possible. Access to and use of this Manual is also provided subject to the acceptance of the Disclaimer set out below.

Disclaimer

About the Manual

This Manual has been drafted by IDS GeoRadar with the due care in order to ensure the accuracy and completeness. However, please note that no warranty is provided about the fact that the Manual will be free from error or omission and about the accuracy and/or fitness to the purpose of any information contained herein.

Any representation and/or statement contained herein has been written in good faith and on the basis that, due to negligence, lack of care or otherwise, IDS GeoRadar is not liable for any loss, including direct, indirect or consequential losses, in contract or in tort for any misstatement or omission contained herein which has occurred or may occur with reference to any representation and/or statement contained in the Manual.

User shall also acknowledge that some information contained in the Manual may not be up to date. IDS GeoRadar may amend the information contained in the Manual from time to time and IDS GeoRadar shall not be deemed liable in the event that damages arise with reference to changed circumstances or other information which may affect the accuracy of any information contained in this version of the Manual.

About the Product

All products designed, produced and distributed by IDS GeoRadar, its Subsidiaries, Affiliated and authorized Distributors (Products) shall not be used, in the design, development, production or use of chemical, biological, nuclear ballistic weapons. Buyer/ User will defend, indemnify and hold IDS GeoRadar harmless against any liability (including attorney's fees) for non-compliance with the terms of this paragraph.

If IDS GeoRadar requires that Buyer/User shall carry out a training with reference to some Product categories, no operation or use of the Products shall be started before its designated Operator/s has got the User Certificate, as defined by IDS GeoRadar specific procedure which the Buyer confirms to know and accept.

For Products which include specific "Operational" software with automatic data processing and analysis "Tools", e.g. RockSpot, IBIS Products; Hydra Products, User shall be aware that the results provided by these "Tools" may be not error free. User that completely relies on the outcomes provided by these Tools only, does it at his own risk.

In no event IDS GeoRadar shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the missed or incomplete observance of the instructions and prescriptions for the use of the Products, either separately or in combination with other products.

This paragraph provides only a non-exhaustive summary of the full Product disclaimer. Please see the full Product disclaimer included into the relevant section of this Manual for detailed information.

Confidentiality

This Manual is strictly confidential and may be subject to legal and other privilege. IDS GeoRadar specifically reserves its rights in relation to this Manual. None of these rights or any other rights subsisting in the Manual are waived if you have received this Manual in error or without permission. No parts of this Manual may be copied, translated or reduced to any electronic medium, or machine-readable form, nor photographed or recorded in any way, without the prior written consent of IDS GeoRadar.

INDEX

1	INTRODUCTION	5
1.1	ESTIMATED READING TIME	5
1.2	PURPOSE	5
1.3	APPLICATION FIELD	5
1.4	INTENDED READERSHIP	5
2	HOW TO READ THE MANUAL	6
2.1	MANUAL LAYOUT.....	6
2.2	SYMBOLS	6
2.3	GLOSSARY & ACRONYMS.....	7
2.4	REFERENCE	7
2.4.1	<i>Bibliography</i>	7
3	SAFETY DIRECTIONS	8
3.1	INTENDED USE.....	8
3.2	REASONABLY FORESEEABLE MISUSE.....	8
3.3	RESPONSIBILITIES	8
3.4	CAUTIONARY NOTES	8
3.4.1	<i>Training</i>	8
4	GENERAL DESCRIPTION OF THE HW	10
4.1	STREAM DP.....	10
4.2	CHASER XR	11
4.3	OPERA XR	11
4.4	DS4000	12
5	UMAP-LOGGER ACQUISITION SOFTWARE	13
5.1	PC SYSTEM REQUIREMENTS	13
5.2	HOW TO INSTALL UMAP-LOGGER.....	13
5.3	DEFINITIONS AND DATA FLOW.....	14
6	HOW TO WORK ON SITE WITH UMAP-LOGGER	16
6.1	HOW TO CONNECT UMAP-LOGGER: <i>AUTO DETECT SYSTEM</i> OR <i>MANUAL SYSTEM SELECTION</i>	16
6.1.1	<i>How to connect uMap-Logger and CHASER XR in MANUAL mode</i>	16

6.1.2	How to connect uMap-Logger and STREAM DP in MANUAL mode.....	19
6.1.3	How to connect uMap-Logger and DS4000 in MANUAL mode	20
6.2	HOW TO USE UMAP-LOGGER	21
6.2.1	How to check the radar Status and settings	22
6.2.2	How to select the trigger type (Encoder or Free Running).....	25
6.2.3	How to select and use Positioning Systems (GNSS, TPS, Grid, Quick Scan)	28
6.2.4	How to change Preferences	33
6.2.5	How to update SW and FW	35
6.2.6	How to Start a New Project	36
6.2.7	How to insert targets.....	38
6.2.8	How to export targets.....	39
6.2.9	How to Create a Project Report	41
6.2.10	How to use Remote Diagnostics (Re.D.)	42
6.2.11	How to work with uMap-Logger Advanced (Optional).....	43
	CDS by the Office	46
	CDS by the Field	47
7	FREE RUNNING MODE: SIDE NOTES	50
8	UMAP-LOGGER: SUMMARY OF THE WORKFLOW	52
9	SYSTEM'S REFERENCE POINT ON B SCAN	54
10	IDS GEORADAR SUPPORT	57
10.1	MYWORLD.....	57
10.2	HOW TO MANAGE RETURN FOR SERVICE	57

1 INTRODUCTION

This document describes **uMap-Logger**, the software that allows to run GPR surveys with **Stream DP**, **Chaser XR**, **DS4000**, and **Opera XR** that are GPR systems. For a full description of these systems (including their parts description and set up), the Users should refer to each specific User's manual.

This document refers to the concepts the user should learn before initiating the utilization of these devices. Therefore, it is mandatory to carefully reading the entire document before starting and operating the systems.

1.1 Estimated Reading Time

The estimated reading time of the entire document is 90 minutes.

1.2 Purpose

The purpose of this document is to drive the Reader through a list of step-by-step actions on the data acquisition SW to run a survey with **Stream DP**, **Chaser XR**, **DS4000**, and **Opera XR** in a thorough and safe way.

The structure of the manual is organized in a collection of "how-to" as well as descriptive chapters. Each chapter contains illustrations and basic descriptions of the main operations to be carried out with the system, to run a full survey.

1.3 Application field

The main application for the aforementioned systems is the collection of GPR surveys, over variable size areas, for underground utility mapping and pipe detection, or other environmental, geological, forensic, archaeological, engineering, or geotechnical applications. Also, any type of buried objects or structures may be the aim of the systems if these are buried within their depth of reach.

1.4 Intended readership

The intended Reader(s) of this manual is(are) the technician(s) in charge of using the IDS GeoRadar GPR systems, that has undergone the training for the use of **Stream DP**, **Chaser XR**, **DS4000**, and **Opera XR**.



Note: All the SW functions described hereinafter that refer to Leica DS4000, also applies to IDS GeoRadar Opera XR

2 HOW TO READ THE MANUAL

2.1 Manual Layout

It is strongly recommended to go through this manual thoroughly, and to read it carefully. However, as the document is divided into a first part that is descriptive, but also into procedures (*How to...*) that are self-consistent, and that can be read sparsely, it is not mandatory to read this document chapter after chapter, in case of need.

Please, note that images and pictures shown in the manual aims at supporting the Users in understanding the operation described. However, pictures and images portrayed herein may, from time to time, slightly differ from the actual Hardware and Software they describe.

2.2 SYMBOLS

Warning messages are an essential part of the Safety Concept of the instrument. They appear wherever hazards or hazardous situations can occur.

Type	Description
 DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a potentially hazardous situation or an unintended use which, if not avoided, could result in death or serious injury.
 CAUTION	Indicates a potentially hazardous situation or an unintended use which, if not avoided, may result in minor or moderate injury.

Warning messages:

- make the user alert about direct and indirect hazards concerning the use of the product.
- contain general rules of behaviour.

For the user's safety, all safety instructions and safety messages shall be strictly observed and followed. Therefore, the manual must always be available to all Users performing any tasks described herein.

DANGER, WARNING, CAUTION and NOTICE are standardized signal words for identifying levels of hazards and risks related to personal injury and property damage.

For your safety it is important to read and fully understand the table below with the different signal words and their definitions. Supplementary safety information symbols may be placed within a warning message as well as supplementary text.



Note text/to keep in mind



Tip information

2.3 Glossary & Acronyms

The following acronyms and naming are used in this manual:

- **CDS:** Cloud Data Sharing.
- **Controller:** PC / Tablet used by the user for data acquisition. The data acquisition SW (client) is installed on the controller.
- **Free Running:** acquisition mode triggered by the “time” not by encoder (wheel or any other type).
- **GNSS:** Global Navigation Satellite System.
- **GPR:** Ground Penetrating Radar or Georadar or Ground Probing Radar.
- **GUI:** Graphical User Interface. A graphical user interface, or GUI, is a form of user interface that allows users to interact with electronic devices.
- **NMEA format:** National Marine Electronics Association format.
- **Re.D.:** Remote Diagnostics.
- **SNR:** Signal to Noise Ratio.
- **SFTP:** Secure File Transfer Protocol.
- **Stream DP:** IDS GeoRadar GPR massive Array System.
- **Chaser XR:** IDS GeoRadar GPR System.
- **DS4000:** Leica Geosystems GPR system
- **Opera XR:** IDS GeoRadar GPR System
- **UBLOX:** GNSS device within the antenna case.
- **uMap-Logger:** IDS GeoRadar software for GPR data acquisition.

2.4 Reference

2.4.1 Bibliography

[BD1] MNG_2022-008 rev 1_2 uMap 1_2

[BD2] MNG_2023-009 rev 1_0 uMap-Logger 1_0

3 SAFETY DIRECTIONS

The following directions enable the person responsible for the product, and the person who uses the equipment (HW and SW), to anticipate and avoid operational hazards.

The Responsible Person for the product, must ensure that all Users understand these directions and adhere to them.

3.1 Intended Use

The intended use is the GPR data acquisition with **Stream DP**, **Chaser XR**, **DS4000** and **Opera XR** systems, for the detection of underground targets, such as:

- Water pipes
- Gas pipes
- Sewers
- Fiber Optic
- Electric cables
- Telephone cables
- Manholes
- Cavities
- Structures in general



NOTE: The product is intended for professional use only; not for consumers use.

3.2 Reasonably Foreseeable Misuse

Below, a list of possible misuse of the system (HW and SW) to be avoided:

- Breaking traffic law.
- Use of the product without instruction.

- Use outside of the intended use and limits.
- Modification or conversion of the product.
- Use after misappropriation.
- Use of products with obvious damage or defects.
- Use with accessories from other manufacturers without the prior explicit approval of IDS GeoRadar s.r.l.
- Inadequate safeguards at the working site.

3.3 Responsibilities

Manufacturer of the product: IDS GeoRadar s.r.l. is responsible for supplying the product, including the user manual and original accessories, in a safe condition.

Person responsible for the product: the person responsible for the product has the following duties:

- to understand the safety instructions on the product and the instructions in the user manual,
- to ensure that it is used in accordance with the instructions,
- to be familiar with local regulations relating to safety and accident prevention,
- to inform IDS GeoRadar s.r.l. immediately if the product and the application becomes unsafe,
- to ensure that the national laws, regulations, and conditions for the operation of electromagnetics transmitters are respected.

3.4 Cautionary Notes

3.4.1 Training

For the safe and proper use of the system, it is essential that it is only operated by skilled users. A skilled user is someone who has successfully completed appropriate training. It is therefore strongly recommended that

all users complete such training from IDS GeoRadar. IDS GeoRadar is not liable for any damages caused by unskilled and/or untrained users.

4 GENERAL DESCRIPTION OF THE HW

uMap-logger is the IDS GeoRadar data acquisition software allowing to run surveys with the following IDS GeoRadar systems:

1. Stream DP,
2. Chaser XR (Including optional Cart, Drag Kit and Positioning Kit),
3. Opera XR
4. DS4000

In this paragraph, a short description of the above systems is presented. For a full description and their set up instructions, the User must refer to each specific User's Manual.

4.1 STREAM DP

Stream DP (Fig. 1) is a GPR system composed of a multi-channel antenna case (named GPR AE32HV) fixed to a compact and manoeuvrable frame.

Stream DP is designed for high productivity and ease of use: light-weight structure for maximal transportability, hot swap batteries for a non-stop performance and multi-environment hardware configuration (rugged and asphalt).

The antenna module emits GPR signal in two different orientations at the same time (Polarization HH and VV). The general idea behind the design of Stream DP systems is to detect the main underground services along the road and their connections, by fully covering the area to be investigated. For this reason, a double polarization GPR antennas array has been designed and implemented, one with transversal polarization (HH, with respect to the scan direction), and one with a longitudinal polarization (VV).

The antenna integrates the radar control unit (FCU, Fast Control Unit) that manages all the input and output. It is directly connected to the laptop and powered by two Li-Ion batteries. The system can be used in two

configurations: 1) **rugged**, mounting four side wheels, suited for difficult terrains; 2) **asphalt**, in which the rear wheels are pivoting.

The controller laptop can be fixed on the rudder, through the tablet holder.



Fig. 1 Stream DP

4.2 CHASER XR

Chaser XR GPR system (Fig. 2) is designed to detect and locate underground features for a wide range of possible applications including, but not limited to Geology, Geotechnical, Environmental, Glaciers, Rivers, Archaeology, Forensic, buried structures in general. It provides georeferenced underground georadar maps if a supported positioning system is used.

The Chaser XR antenna integrates the radar control unit (FCU, Fast Control Unit) that manages all the input and output. It is directly connected to the laptop either via WiFi or via LAN cable and powered by two Li-Ion batteries. The system (antenna) can be used in four configurations: **Chaser XR Lite** (Basic pack), **Rugged** (by using the Optional *Drag Kit* for rough terrain), **Positioning Kit** (by using the Optional Positioning Kit for GNSS and/or TPS), and with **cart**.

A full description of all configurations is available on the Chaser XR User's manuals and quick guide.



Fig. 2 Chaser XR Lite

4.3 Opera XR

The **Opera XR** is the entry level GPR system by IDS GeoRadar srl (Fig. 3); it is designed to detect and locate underground features for a wide range of possible applications including, but not limited to utility mapping, geology, geotechnical, environmental, glaciers, rivers, archaeology, forensic, buried structures in general. It provides georeferenced underground maps if a supported positioning system is used and correctly paired to **Opera XR**.

The IDS GeoRadar **Opera XR** Utility Detection system includes the following packages:

1. **Opera XR**,
2. **Opera XR GNSS Support Kit** (Optional),
3. **Opera XR Off-Road Wheels Kit** (Optional).



Fig. 3 IDS GeoRadar Opera XR

4.4 DS4000

DS4000 is the entry level GPR system by Leica Geosystems; it is designed to detect and locate underground features for a wide range of possible applications including, but not limited to utility mapping, geology, geotechnical, environmental, glaciers, rivers, archaeology, forensic, buried structures in general. It provides georeferenced underground maps if a supported positioning system is used and correctly paired to DS4000.

The Leica Geosystems **DS4000** Utility Detection system includes the following packages:

- a. **DS4000 starter kit** (Fig. 4),
- b. **DS4000 surveyor Kit** (Fig. 5).



Fig. 4 Leica DS4000 Starter Kit



Fig. 5 Leica DS4000 Surveyor Kit

5 UMAP-LOGGER ACQUISITION SOFTWARE

5.1 PC system Requirements

uMap-Logger is the data acquisition software to be installed onto the PC that allows the manage the GPR data acquisition with Stream DP, Chaser XR, DS4000. The technical specifications of the PC where to install uMap-Logger are detailed below:

- **Operative System:** Windows 10 64 bit (required)
- **CPU:** Intel Core i5-8365U @1.60GHz (or superior)
- **Graphics:** Intel® UHD Graphics 620 (or superior)
- **Memory (RAM) - 2 slots upgradable:**
 - Performance model: 16GB up to 32GB total (DDR4-2133MT/s)4
 - 8GB (required)
- **Storage Drive Shock-mounted with Heater (quick-release):**
 - 256GB SSD (required)
- **Wi-Fi card:**
 - 802.11n Wi-Fi 4 (required)
 - 2nd card 802.11n (recommended)
- **USB:** 1 USB 3 port (required)
- **Ethernet Card:** Gigabit Ethernet (recommended)

5.2 How to install uMap-Logger

Once the latest version of uMap-Logger has been downloaded from the *myDownload* area of *myWorld* (<https://myworld-portal.leica-geosystems.com>), the User may proceed with the installation procedure as detailed below.

To install uMap-Logger software:

1. open the installer folder and “Run as Administrator” the application file (Fig. 6).

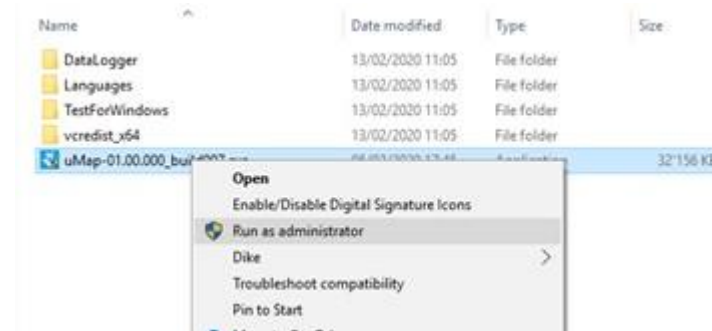
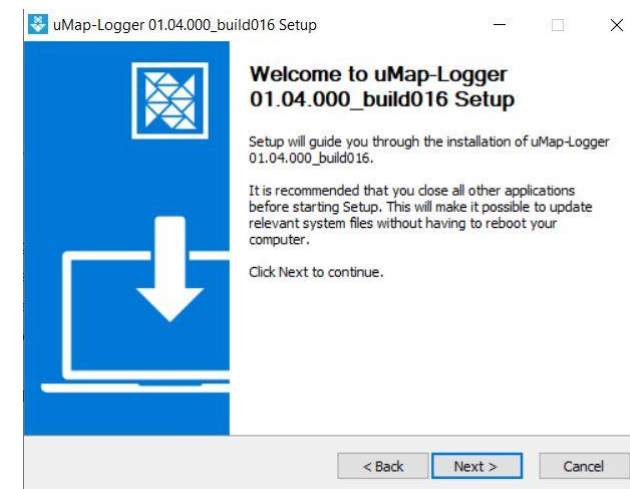


Fig. 6 uMap-Logger Installation

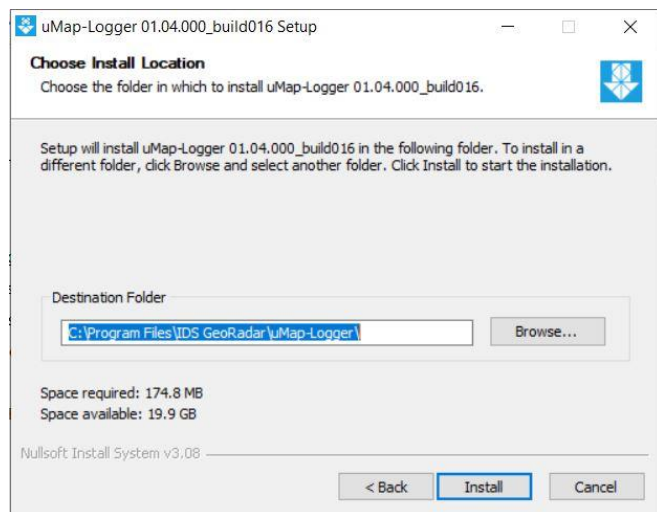


Note: It is important to always run the latest uMap-Logger SW version that can be downloaded from myWorld website.

2. Then follow the (4) steps indicated in the windows appearing during the software installation wizard as portrayed in the images sequence of (Fig. 7). Press “Next”,



3. Then, choose the installation Location (Directory) where to instal the SW on the field PC,



4. Read and accept (press “I Agree”) the **Hexagon Geosystems Software License Agreement**,



5. Press “Finish”, to complete installation.

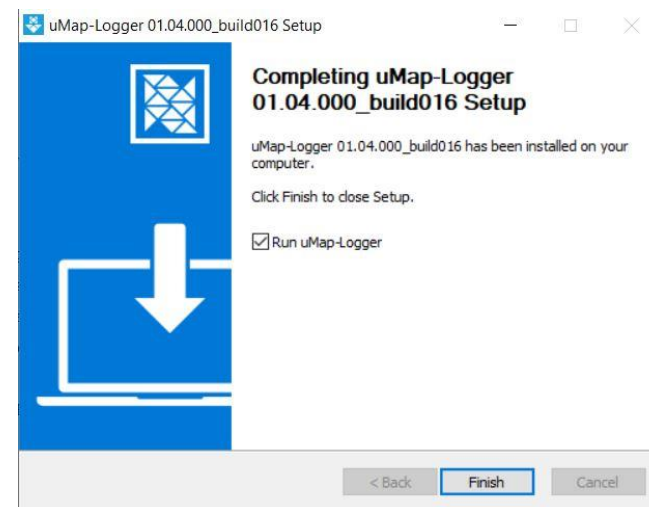


Fig. 7 uMap-Logger Installation sequence

By the end of the installation procedure the uMap-Logger shortcut will be displayed on the PC Desktop.



5.3 Definitions and data flow

Before starting to illustrate the workflow implemented in uMap-Logger, some concepts, and definitions to increase the user’s awareness in handling the workflow itself, are illustrated below:

- **Working Folder:** is the folder created on the Controller PC, where the acquisition SW is installed and runs. The working folder contains all the information that allows the User to review a mission, including Logs, Projects and Settings. It is normally located in the directory C:\Documents\UMap.
- **Data Folder:** is the project folder created in the memory of the on-board PC. It Contains:
 - The raw data.

- The positioning system datum.
- Project info file.
- **Project:** It can consist of one or multiple surveys. The different surveys that make the project cannot include different HW configurations. Multiple surveys may be generated in a single Project in the case of Resuming and restart working on a previous Survey.
- **Survey:** Single GPR data set collected after Starting a new Project. Each survey can only contain consistent swaths between each other (such as configuration and positioning system).
- **Swath:** Each single set of GPR data acquisition line between the action of pressing “Play” and “Stop” during a survey acquisition.
- **Resume:** How the user can continue to acquire an open project in review mode. The resume operation generates a new survey if the acquisition configuration changes.

The **scan step** is defined as the measure of the distance between every consecutive radar trace displayed in the radargrams.

Stream DP, Chaser XR, DS4000 and Opera XR GPR emits more than a single georadar trace in a single scan step, depending on the system velocity, and by performing the average of these traces, it is possible to increase the SNR (Signal to Noise Ratio). This sampling and averaging are the so-called **Dynamic Stacking**.

6 HOW TO WORK ON SITE WITH UMAP-LOGGER

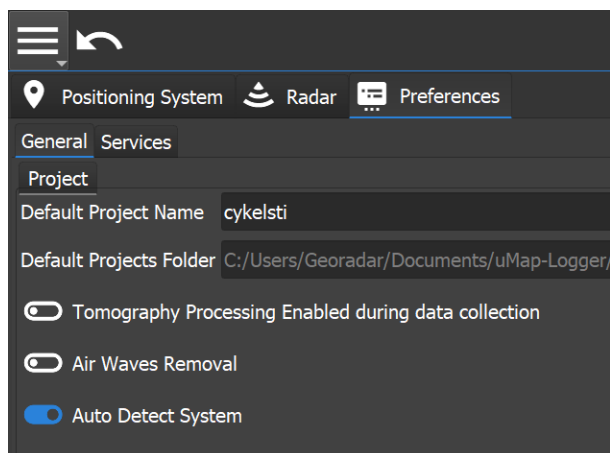
This section of the User's Manual illustrates, in a schematic way, the main functions that can be implemented in uMap-Logger to correctly run Stream DP, Chaser XR and DS4000 surveys.

6.1 How to connect uMap-Logger: *Auto Detect System* or *manual system selection*

The first step to run the **Stream DP**, **Chaser XR**, and **DS4000** via uMap-logger is to pair the PC where uMap-Logger is installed, and the antenna. To do this, and before starting uMap-Logger software, the User must connect the PC to the LAN port, using an RJ45 cable. The Ethernet port must be in DHCP mode (dynamic address).

Only for Chaser XR, the connection can be done both WiFi and via LAN cable.

As uMap-Logger is launched, under the **Preferences menu**, it is possible to select the **Auto Detect System** mode: this function allows the SW to automatically detect the system connected once the HW is turned on.



Alternatively, the selection of the HW, can be manually selected as illustrated below.

6.1.1 How to connect uMap-Logger and CHASER XR in MANUAL mode

The connection between the field PC (where the uMap-Logger data acquisition SW is installed) and the Chaser XR antenna can be fulfilled 1) via WiFi or, 2) via LAN Cable.

Regardless the specific field configuration to be used to collect GPR data, the common steps to start up the system before commencing any data acquisition on site are:

1. **Connecting Chaser XR antenna and field PC in Wi Fi Mode:**

- a. Open the battery compartment on top of the antenna case and insert the (2) batteries. Then close the battery compartment. Note: The system can be operated with 1 battery only, in case of need.



- b. Press the O/I button beside the battery compartment to start up Chaser XR.



- c. 3 leds are present on top of the antenna case: 2 indicating the "battery" status, and 1 the "System" status. As the system is turned on and the batteries are fully charged, the battery led shall be solid green (red, in case battery level is low) and the "system" led shall have long-paused blinking

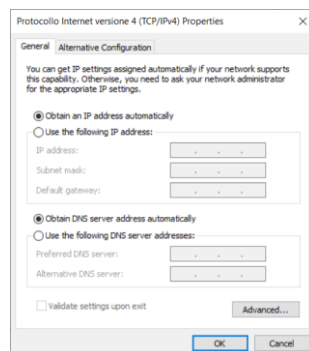
green-to-blue for three times, indicating that the internal WiFi module is starting up to generate its own local network.



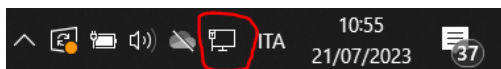
- d. As the “System” led turn to solid green, the Chaser XR internal WiFi is active and generating its own network.



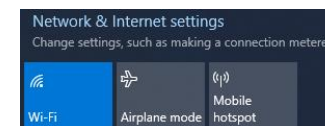
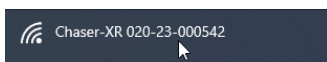
- e. Turn on the field PC where the *uMap-Logger* software is installed (firewall and antivirus must be OFF). The IP address of the WiFi port of the PC must be set to DHCP (Dynamic mode and NOT Static IP address).

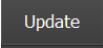


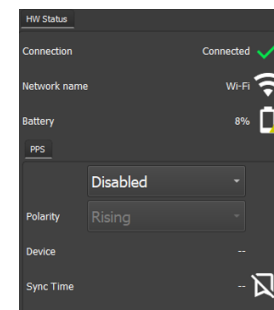
- f. Click on the network icon on the lower right corner (red circle in the image below).



- g. Select the WiFi network generated by the Chaser XR antenna including the serial number of that specific antenna (e.g.: **Chaser XR 020-23-000542**). Wait for connection to be established.



- h. Turn on *uMap-Logger* software, then press on the Radar diagnostic icon , press update , and the connection between the antenna and the PC is established.



2. Connecting Chaser XR antenna and field PC using LAN Cable:

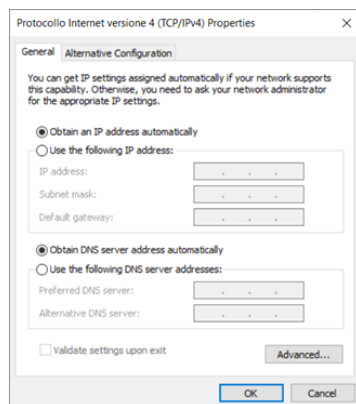
- a. Open the battery compartment on top of the antenna case and insert the (2) batteries. Then close the battery compartment.

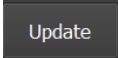


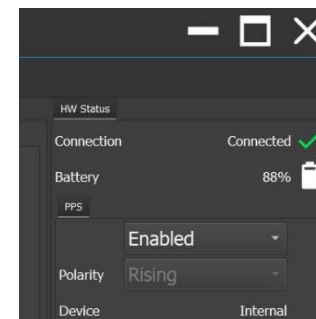
- b. Connect the field PC and the Chaser XR antenna via the LAN cable provided with the system: one end to the antenna LAN connector, and the other end to the LAN port of the PC.
- c. Press the O/I button beside the battery compartment to start up the antenna. In this case, the green-to-blue blinking (described in the previous paragraph) will be not present due to the connection via LAN cable, but the WiFi module will be start up and usable in any case.



- d. Turn on the field PC where the *uMap-Logger* software is installed (firewall and antivirus must be OFF). The IP address of the Ethernet port of the PC must be set to DHCP (Dynamic mode and NOT Static IP address).



- e. Turn on uMap-Logger, press on the Radar configuration icon , press update , and the connection between the antenna and the PC is established.



Chaser XR, is now ready to be used on site for data acquisition.

Occasionally, User may swap connection mode between field PC and Chaser XR antenna from WiFi to LAN cable and vice versa. Also, field PC may have two Ethernet ports.

A possible mismatch between connection type selected on uMap-logger and the actual HW setup, shall lead to connection error. The mismatch (IP Conflict) is automatically detected, and a pop-up message (Fig. 8) shall be visualized asking the User to select the correct Connection type.

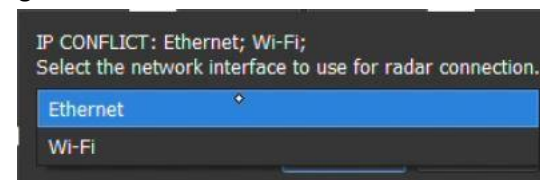


Fig. 8 IP Conflict pop up message

Also, in this case, the HW Status tab shall show this connection error (Fig. 9).

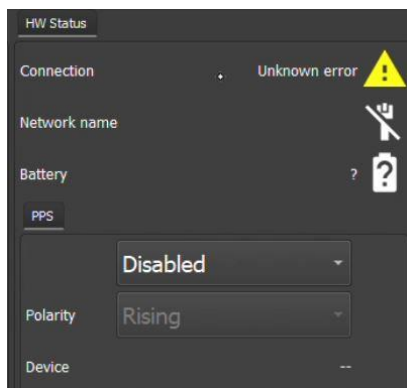


Fig. 9 HW Status Connection Error

Same pop up and HW Status messaging frame appears when the field PC has two Ethernet ports, and the LAN cable connecting the PC and Chaser XR antenna, it is not connected to the correct one.

6.1.2 How to connect uMap-Logger and STREAM DP in MANUAL mode

1. Open the battery compartment on top of the antenna case and insert the (2) batteries. Then close the battery compartment.

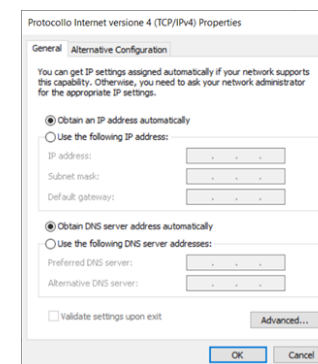


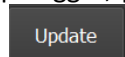
2. Connect the field PC and the Stream DP antenna via the LAN cable provided with the system: one end to the antenna LAN connector, and the other end to the LAN port of the PC.

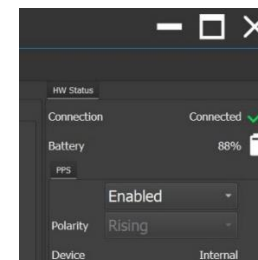
3. Press the O/I button beside the battery compartment to start up the antenna.



4. Turn on the field PC where the *uMap-Logger* software is installed (note: firewall and antivirus must be OFF). The IP address of the Ethernet port of the PC must be set to DHCP (Dynamic mode and NOT Static IP address).



5. Turn on uMap-Logger, press on the Radar configuration icon , press update , and the connection between the antenna and the PC is established.



Stream DP is now ready to be used on site for data acquisition.

6.1.3 How to connect uMap-Logger and DS4000 in MANUAL mode

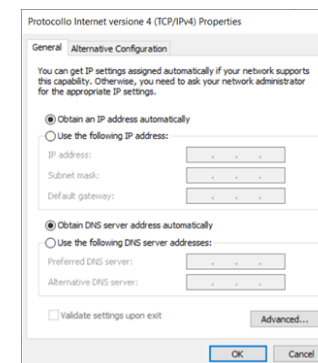
1. Open the battery compartment on top of the antenna case and insert the battery (1 or 2). Then close the battery compartment.

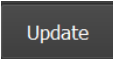


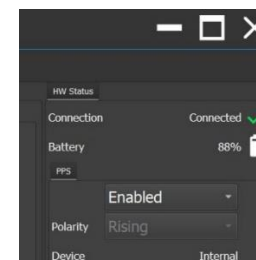
2. Connect the field PC and the system's antenna via the LAN cable provided with the system: one end to the antenna LAN connector, and the other end to the LAN port of the PC.
3. Press the O/I button beside the battery compartment to start up the antenna.



4. Turn on the field PC where the *uMap-Logger* software is installed (note: firewall and antivirus must be OFF). The IP address of the Ethernet port of the PC must be set to DHCP (Dynamic mode and NOT Static IP address).



5. Turn on uMap-Logger, press on the Radar diagnostic icon , press update , and the connection between the antenna and the PC is established.



DS4000 is now ready to be used on site for data acquisition.

6.2 How to use uMap-Logger

As the connection between field PC and the system is established, and uMap-Logger is launched, the Main Page of the software pops up (Fig. 10).

Starting from here, User can control all the aspects of a project, including:

- Running a survey,
- System Diagnostics,
- Positioning system,
- Software interface preferences,
- Software update and version.



Fig. 10 uMap-Logger Main page

Actions that can be run from this page, are:

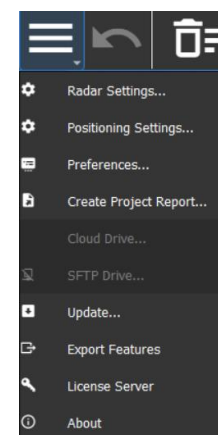
1. To **Create a New Project** by using the icon .
2. To **Open the Projects Explorer** and explore all previous Projects by using the icon . Also, an existing project can be resumed to

continue collecting data on that very site (for example, resuming and restarting collecting data, after a break).

3. **To check** for: Battery level of the system , radar connection status  and positioning system connection .

4. To **Explore Radar and Positioning settings, Preferences, Cloud Drive, SFTP Drive, Update, License Server** and **About** through the

Hamburger menu below ; also **Project Report** and **Features** of an existing Project can be exported.



5. To **go back** to the previous page by using the back-arrow icon .
6. To **Minimize, Expand, Exit** by either selecting one of the following icons .

7. To Link **uMap-Logger Advanced** to the Cloud Drive Providers 

or to the SFTP Drive .

6.2.1 How to check the radar Status and settings

By pressing the icon  or the “hamburger”  menu on the top left corner of the main page, the **Radar Configuration** page pops up. Its aspect and content vary if the **Stream DP** (Fig. 11), **Chaser XR** (Fig. 12) or **DS4000** (Fig. 13) is selected from the Radar Configuration drop-down menu on the top left corner.

Three main panels are available here:

- 1) the **Radar configuration** tab (Fig. 14, Fig. 15, Fig. 16, and
- 2) the **HW Status** (Fig. 17),
- 3) **PPS** tab (Fig. 17).



Note: The Free Running mode and the PPS function are NOT available when in use with DS4000/Opera XR

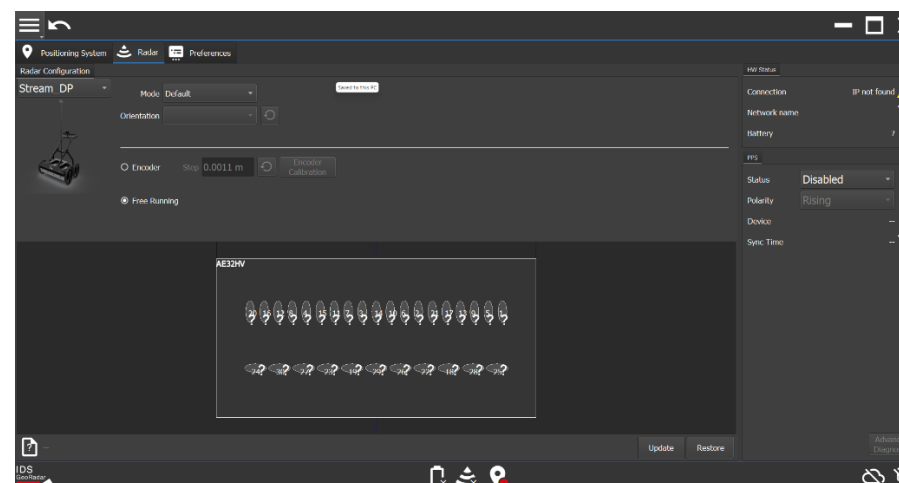


Fig. 11 Stream DP Radar Status and Settings Window

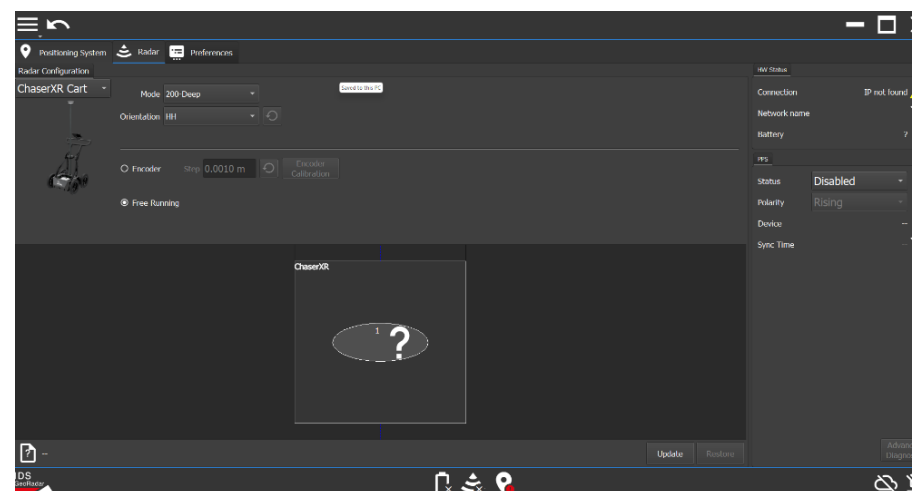


Fig. 12 Chaser XR Radar Status and Settings Window

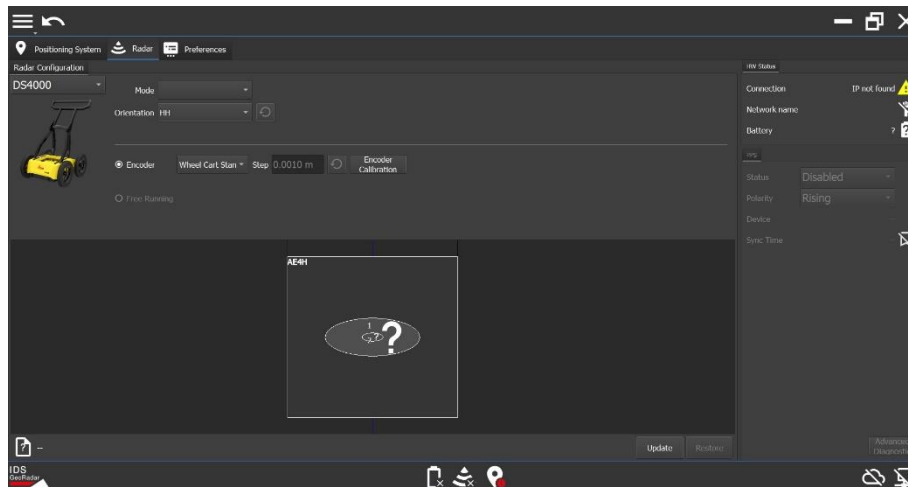


Fig. 13 DS4000 Radar Status and Settings Window

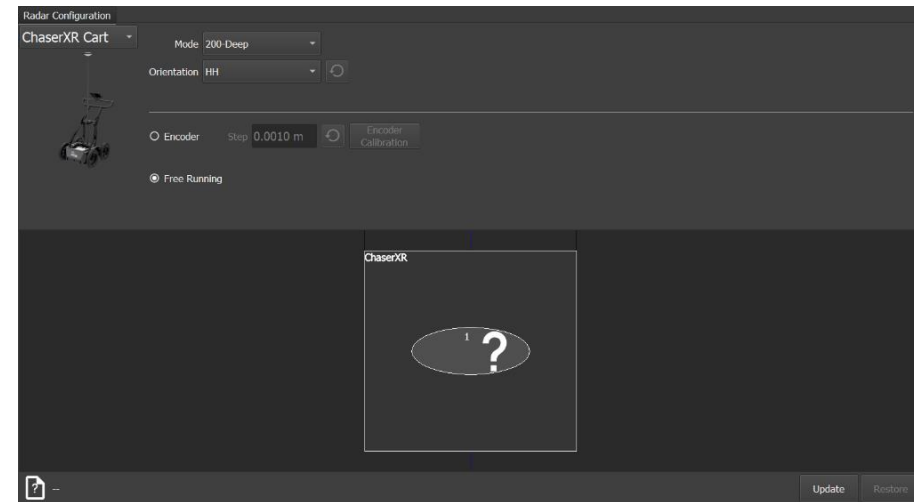


Fig. 15 Chaser XR radar configuration tab

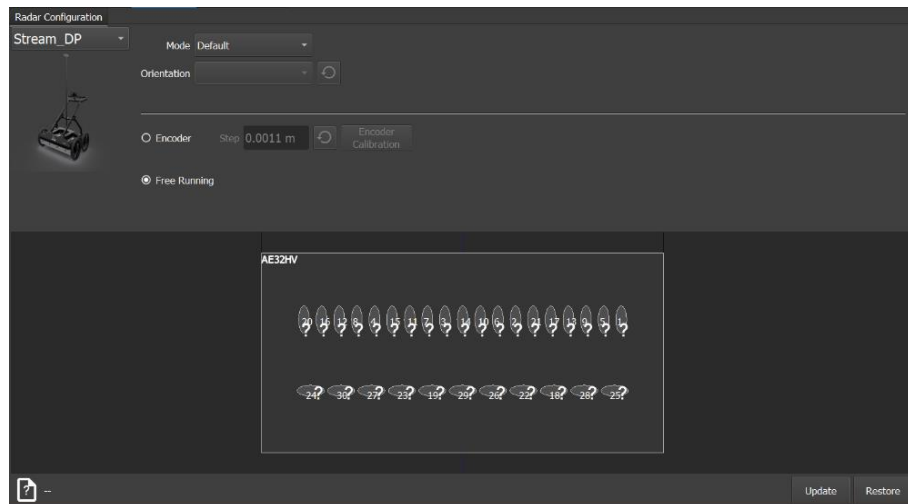


Fig. 14 Stream DP radar configuration tab

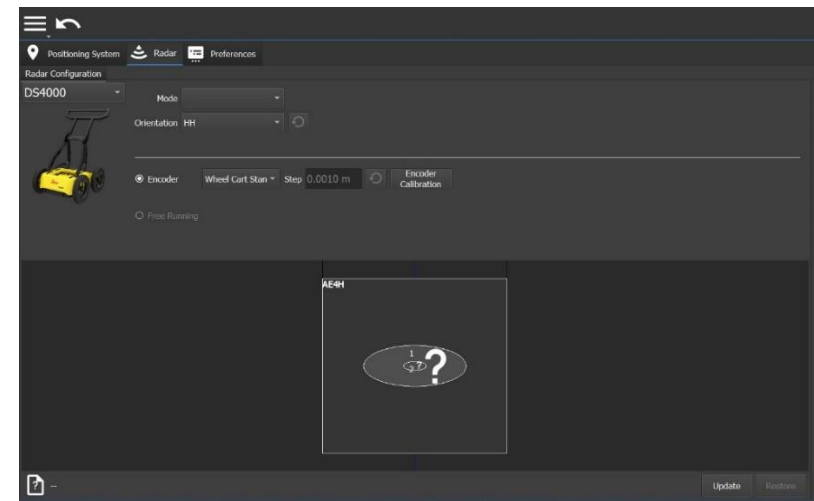


Fig. 16 DS4000 radar configuration tab

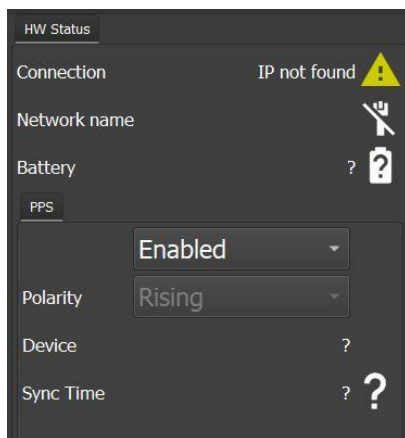
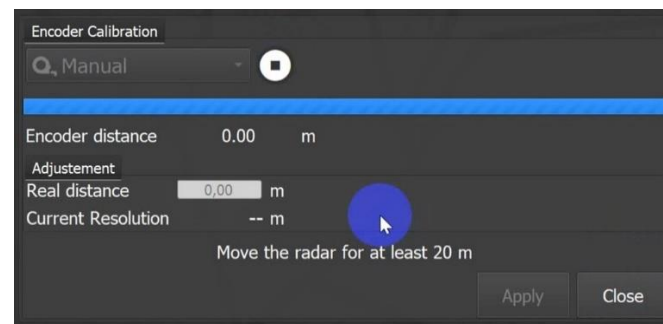


Fig. 17 HW Status and PPS tab

1. In the **Radar configuration** tab, the user can:
 - Select the system type (Stream DP, Chaser XR, DS4000).
 - Select the Mode (only for Chaser XR, 200Deep and 200-900)
 - Select the trigger type:
 - Encoder (signal trigger by space)
 - Free Running (signal trigger by time. NOT AVAILABLE for DS4000)
 - Select the Encoder type (for Chaser XR and DS4000)
 - Verify the status of antennas and channels (
  Connected and Working,
  Not Connected and not working).
 - Vary the antenna orientation HH or VV (only for Chaser XR and DS4000)
 - Vary the encoder step by running the Encoder Calibration by pressing the icon
 .

The encoder Calibration can be run in “Manual” (Fig. 18) or “Positioning System” (Fig. 19) mode. This depends on the fact that a metric tape or the GNSS paired with the system, may be used for the encoder calibration procedure.



Note: PPS tab is NOT active when in use with DS4000

Fig. 18 Encoder Calibration in Manual mode

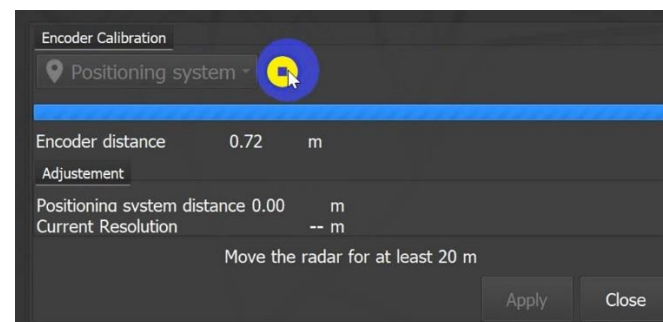


Fig. 19 Encoder Calibration in Positioning system mode

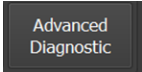
In **MANUAL mode**, the user must click first on the icon , run a known distance (for example 10 meters), click on  and then insert the distance travelled against a metric tape (the actual distance measured).

In **POSITIONING mode**, (**RTK must be active**) the encoder calibration is done by using the GNSS (or the Total Station) position accuracy. The user must click

first on , run at least 20m, click on  and then the new encoder value will be automatically updated.

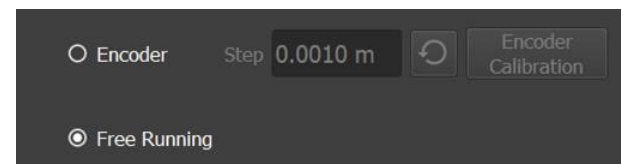
2. In the **HW Status** and **PPS tab** (Fig. 17), the user can check:

- **Connection:** Shows the status of the connection between uMap-Logger and the system,
- **Network name,**
- **Battery:** Shows the status (in %) of the battery, which is connected to the system,
- Enable or disable the internal antenna PPS from drop-down,
- **Device type,**
- **PPS Sync Time:** it displays the last time of synchronization of the PPS signal.

Advanced Diagnostic Logs , can be downloaded by the icon at the lower right corner of the panel.

6.2.2 How to select the trigger type (Encoder or Free Running)

By the Radar configuration tab of **uMap-Logger** (Fig. 14, Fig. 15 , Fig. 16), it is possible to select if the acquisition is triggered by the **Encoder** or by time in **Free Running** mode (by activating the corresponding tick in the image below).



The **Free Running** mode differs from the **Encoder mode** because the latter allows the GPR data trigger through the wheel encoder that is part of the system, whereas for the **Free Running** the trigger is given by the time and disregards the encoder that can be removed or disconnected, for this type of GPR data acquisition.

The Free Running possible applications are:

1. When a wheel encoder would not be in close contact with the ground (e.g.: the GPR system is placed on a boat for river applications), or
2. When wheel encoder would not run correctly on a solid surface (e.g.: the GPR is moved on snow for glacier study applications), or
3. When a positioning system (GNSS or TPS) cannot be used, but reference position markers are needed (e.g.: tunnel applications), or
4. In case of failure of the wheel encoder.

Free Running mode can run with or without GNSS or TPS.

- **Free Running with GNSS or TPS:**

- Select the Positioning system (par. 6.2.3),
- Pair the GNSS or TPS
- Run the acquisition starting a New Project (par. 6.2.6)

- **Free Running in Line or Grid mode:**

- Select the None (Line/Grid) from the Positioning System (par. 6.2.3)
- Draw the reference acquisition Line (par. 6.2.3)

The Free Running acquisition mode in Line or Grid, allows the User to include markers on the B-Scan, when running a survey. Markers allows the User to

have a direct correspondence between specific point(s) on the B Scan and Points of Interest (PoI) present along the acquisition line (e.g.: a spray mark including distance from start of the line, or an object present on site at a known distance with respect to the beginning of the data acquisition line).

Markers can be added:

- **Automatically** (Markers are added automatically at specified distance),
- **Manually** (User includes marker during data acquisition).

To add markers **Automatically**, follow the steps below:

- Select the Free Running tab at the left lower corner of Fig. 20
- Slide the icon  to enable Automatic Markers,
- Enter the expected Speed Value during data acquisition (0,9m/s in Fig. 20)
- Enter the distance between markers (5 m in the example of Fig. 20)
- Start the acquisition.

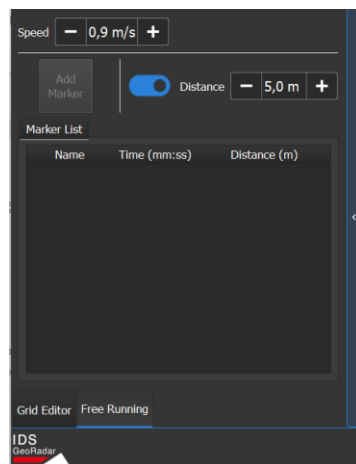


Fig. 20 Add automatic Markers



NOTE: Speed entered in the acquisition panel (Fig. 20) should be as much as possible close to the “real” one. This, affect the progress of the B Scan on screen with respect to the actual position on site.

The B Scan shall appear with markers, similar as in the example of Fig. 21, below.

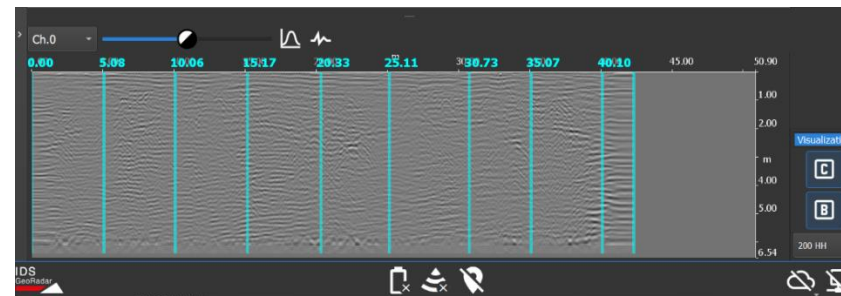


Fig. 21 Example of B Scan with Automatic marker each 5 meter

To add markers **Manually**, follow the steps below:

1. Select the Free Running tab at the left lower corner of Fig. 20,
2. Slide the icon  to disable the Automatic Markers (Fig. 22),

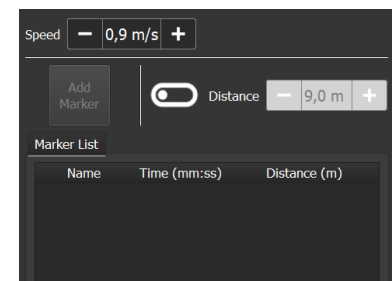
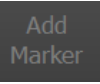


Fig. 22 Add Markers Manually

3. Enter the expected Speed Value during data acquisition (0,9m/s in Fig. 22)
4. Start the acquisition.

5. During data acquisition press the icon  every time the User want to add a marker at a specific location.
The B Scan shall appear with marker as in the example of Fig. 23, below.

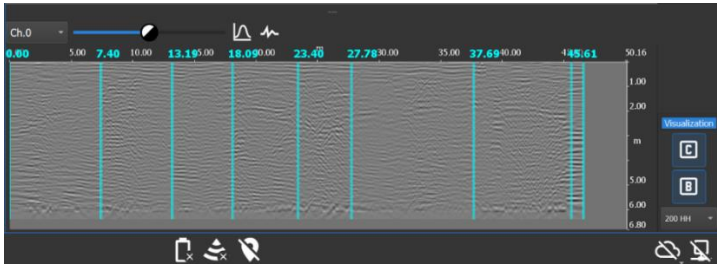


Fig. 23 Add Marker Manually

 **NOTE:** Markers can be applied ONLY in Free Running Acquisitions mode in LINE or GRID. Not with GNSS or Total Station.

In both cases, with or without GNSS or TPS, the list of the added Markers shall appear on the left panel (Fig. 24).

 **NOTE:** Markers can be applied ONLY in Free Running Acquisitions mode in LINE or GRID. Not with GNSS or Total Station.

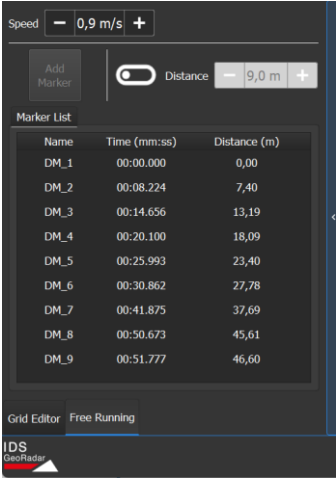


Fig. 24 Markers List

In the following panel (Table 1) are reported a few examples of expected speed value to be used depending on the system and conditions.

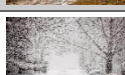
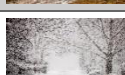
SYSTEM		SURFACE		SPEED (m/s)
Chaser XR 		Paved/Flat		0.9-1
		Grass/moderately rough		0.7-0.8
		Dirt/rough		0.6-0.7
		Soft (like snow or sand)		0.5-0.6
Chaser XR with CART 		Paved/Flat		1-1.2
		Grass/moderately rough		0.8-0.9
		Dirt/rough		0.7-0.8
		Soft (like snow or sand)		0.4-0.5
Stream DP 		Paved/Flat		1-1.4
		Grass/moderately rough		0.8-0.9
		Dirt/rough		0.5-0.75
		Soft (like snow or sand)		0.2-0.4

Table 1 Speed of acquisition (Average Values). The above values are “indicative” and may differ based on actual surface condition.

Values portrayed herewith should be considered only as indicative. Speed value refers to the velocity with which the operator is moving when operating the system.

6.2.3 How to select and use Positioning Systems (GNSS, TPS, Grid, Quick Scan)

By pressing the icon  or the “hamburger” menu  on the top left corner of the main page, the Positioning System page pops up (Fig. 25).

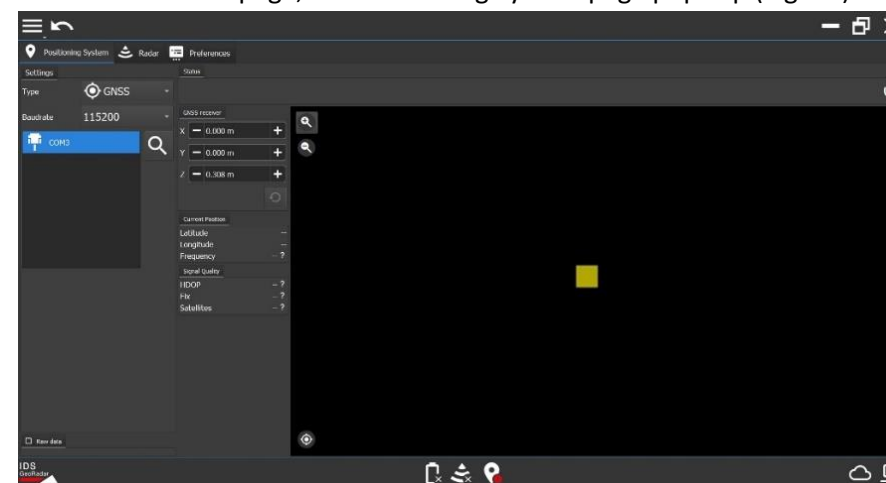


Fig. 25 Positioning setting window

By the “Type” drop down menu (Fig. 26), the User may select among the following positioning system to be paired with the systems:

- GNSS
- TPS
- Grid
- Quick Scan

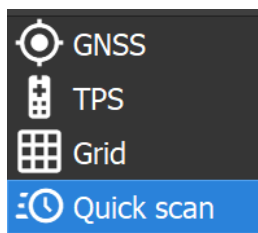
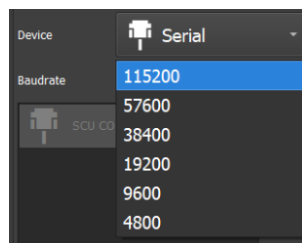


Fig. 26 Positioning Types

1. For both, the **GNSS** and **TPS mode**, User can:

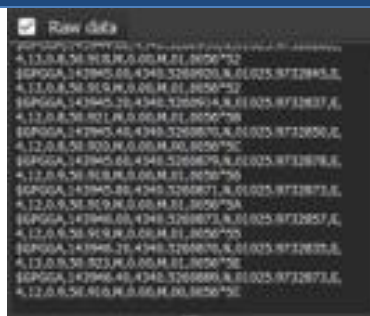
- Check and change the Baudrate in case that serial COM port is used.



- Verify the NMEA data streaming in real-time.



NOTE: The correct format for GNSS data is NMEA GGA with 5 Hz refresh rate. For TPS also a Pseudo NMEA strings in imperial system is accepted.

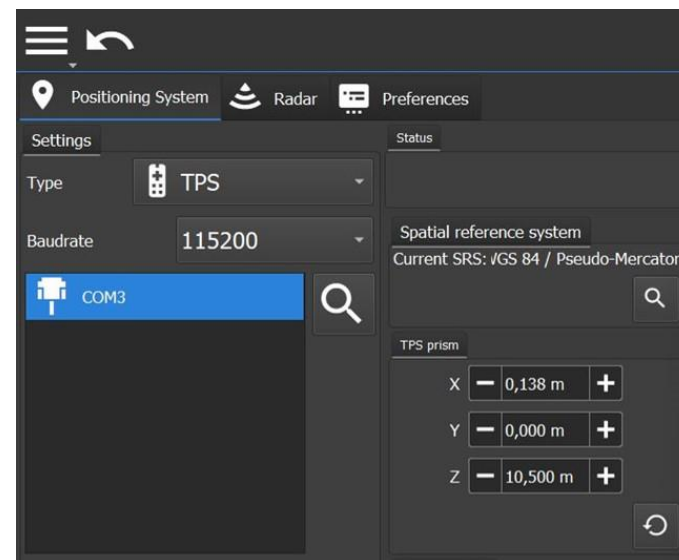


- Check for the GNSS or TPS **Current position** and **Signal Quality**, by their respective window, below.

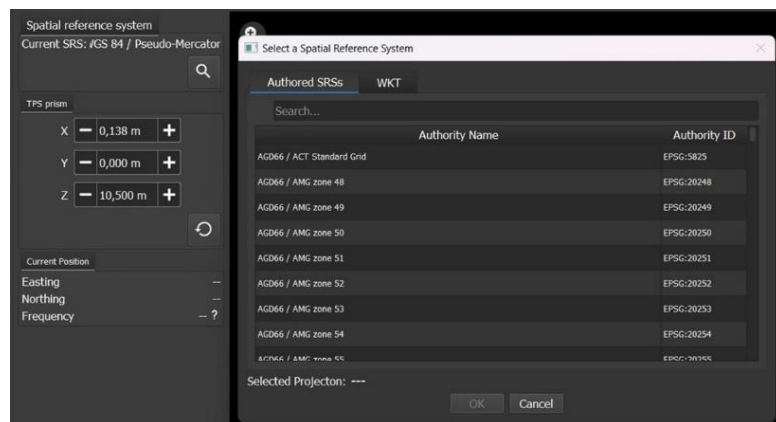
Current Position		Signal Quality	
Latitude	432888 E	HDOP	0.90 ✓
Longitude	675435 N	Fix	RTK Fix ✓
Frequency	5 Hz ✓	Satellites	13 ✓

2. For **TPS Mode** User may select Local Coordinate or a specific Projection by selection from the Spatial Reference System (SRS) library available:

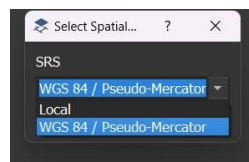
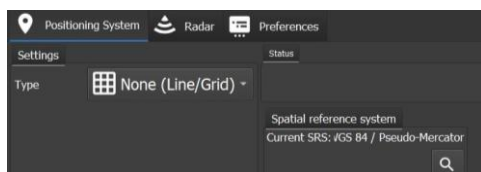
- Select TPS mode and press the icon  under Spatial Reference System,



- Search and select the desired SRS by the list,



- Press OK.
3. When “**Grid**” Positioning mode is selected, User can run acquisition when neither a GNSS nor a robotic Total Station are available. Data acquisition in Grid mode, regardless the HW, can be designed by the following procedure:
- Select **Grid**, by the Positioning Type drop down menu (Fig. 26),
 - Press the  icon under Spatial Reference System to define if the Line/Grid is going to be **Local** (in meter) or in **WGS84/Pseudo Mercator**,



NOTE: Internet Connection must be available on the field PC for the background map to show.

- Press the  icon on the top left corner,
- Enter the New Project window and press  to start editing a new Line/Grid (Fig. 27).

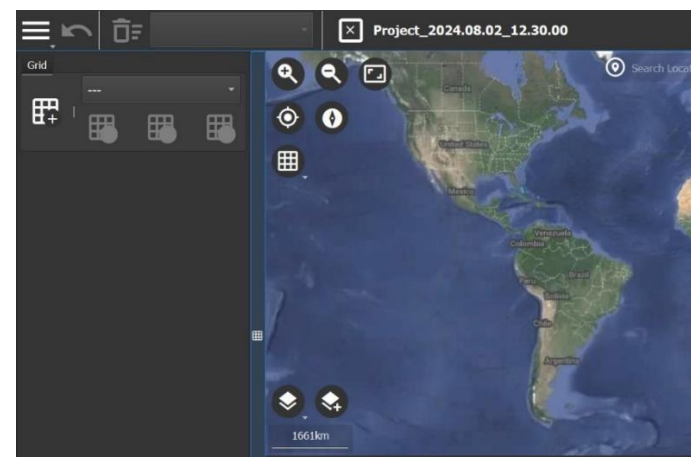


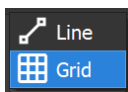
Fig. 27 Grid Baseline generation

- Edit the **Grid** or **Line** (in this example the “Baseline” of a GRID) by:
 - Selecting the Location where to carry out the survey, by the search location toolbox 

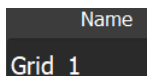


TIP: Enter first the ZIP code and name of the area, to make the search easier.

- Specifying if it is Line or Grid



- Defining the Name **Grid 1**



- Adding, on the area map, the first point of the Baseline by pressing (white, squared dot in Fig. 28).

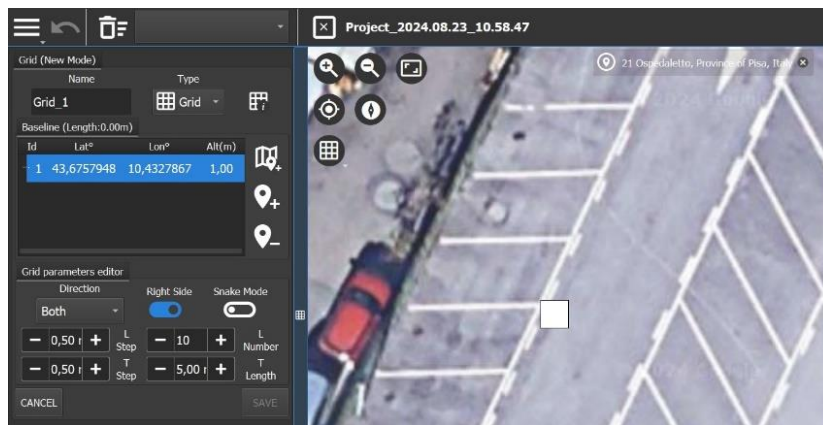


Fig. 28 adding first point to grid baseline

- Adding the second point of the Base line by the same icon shall automatically generate a grid (only longitudinal lines if Stream DP is selected, Fig. 29).

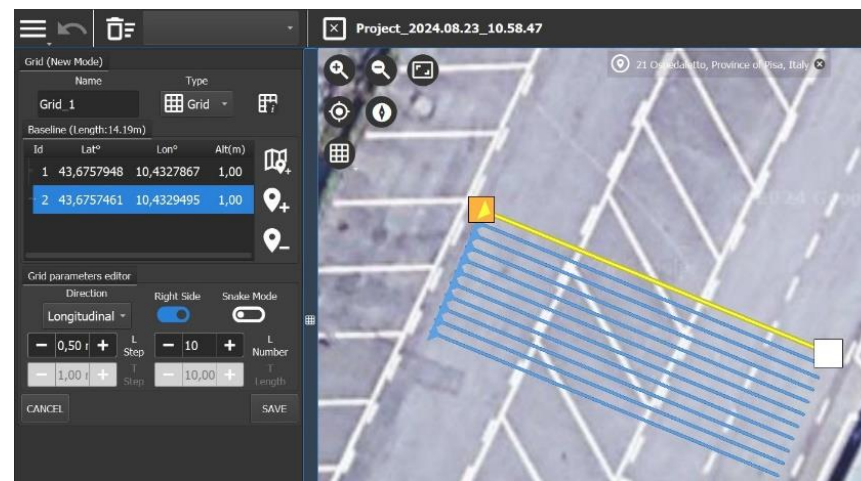


Fig. 29 adding second point to grid baseline

- Once the Baseline (and the Grid) is defined (Fig. 31), it can be edited (Fig. 30), by the following:

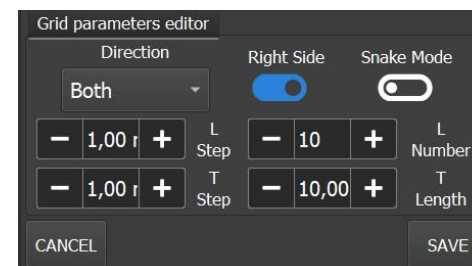


Fig. 30 Edit the VT-Grid

- Defining the direction of acquisition of the Grid line if Longitudinal, Transversal, or Both directions,
- Defining if the Grid Lines are on the Right or Left side of the baseline

- Defining whether the acquisition lines are all in the same direction or in “snake” mode ,
- The Number of Lines, Length, and Offset, can be customized by the toolbox below.

-	1,00	+	L	-	10	+	L
			Step				Number
-	1,00	+	T	-	10,00	+	T
			Step				Length

SAVE

Once designed, the acquisition Line or Grid can be saved by the button.

The Grid or Lines so defined, shall be saved in the project list, and can be recalled at the start of the survey.

In the example below, a grid of 15 x 10 meters with 1 meters longitudinal and transversal offset, has been created (Fig. 31).

Additionally, the following action can be run:

- Press  to add a new Grid/Line.
- Press  to delete the current Grid/line.
- Press  to visualize Grid/Line details.
- Press  to edit the current Grid/Line.

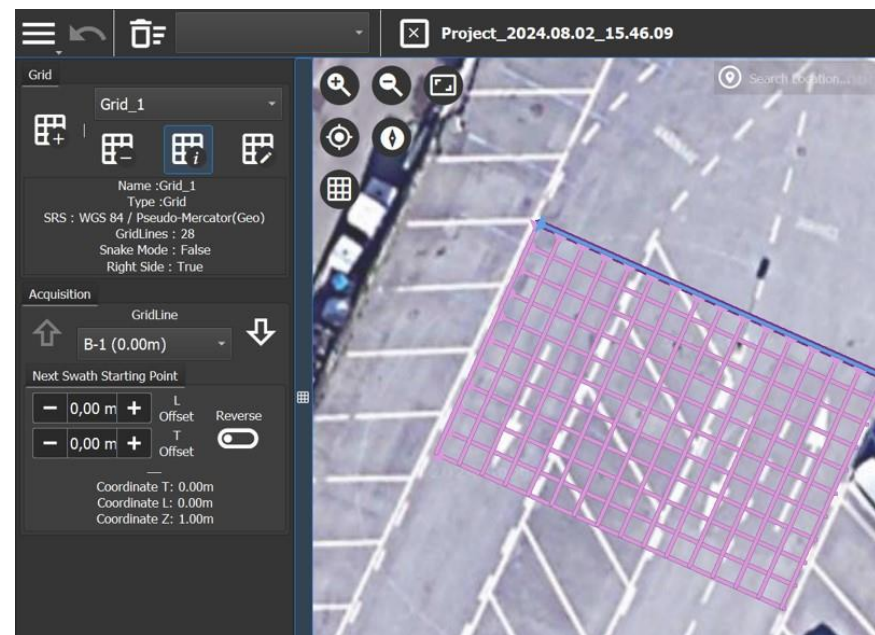


Fig. 31 Grid Parameters Editor

Swath Starting Point can be edited (for example because of the presence of obstacles), by varying the **L Offset** and **T Offset** after double-clicking a single

line of the Grid/Line, to be edited ().

Pressing the icon  enables the polyline editor, allowing to generate a curved acquisition grid or line (Fig. 32).

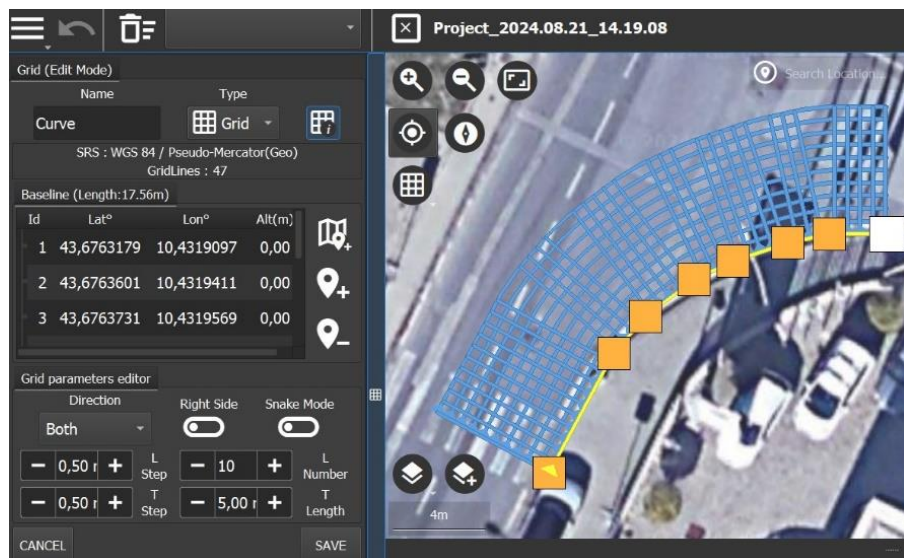


Fig. 32 Polyline editor

Besides the pairing of different types of positioning systems, Chaser XR and DS4000 allows for deploying of variable HW configurations (e.g.: HH or VV Polarization, 200deep or 200-900MHz configurations, and combinations of these). If, on a current Project, a different HW configuration with respect to the one in use, is “Resumed” the error message below (Fig. 33), would pop up.

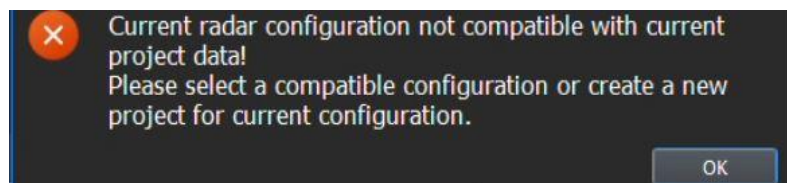


Fig. 33 Radar Configuration mismatch

6.2.4 How to change Preferences

By the **Preference** Icon  under the Hamburger menu , the User shall be prompted to the Preferences window and find two tabs, namely (Fig. 34):

- **General** and,
- **Services**

➤ Through the **General** tab User may Customize the following:

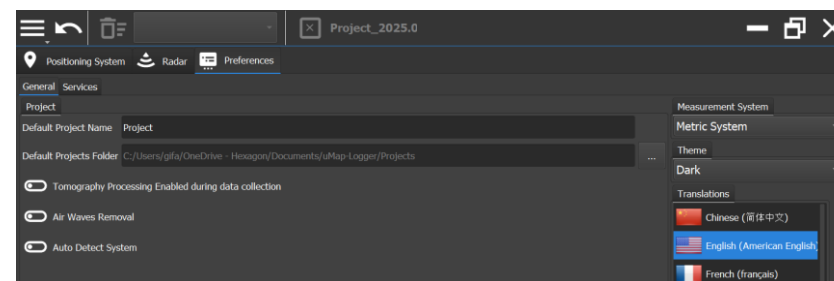
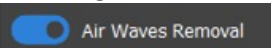


Fig. 34 Preference Window

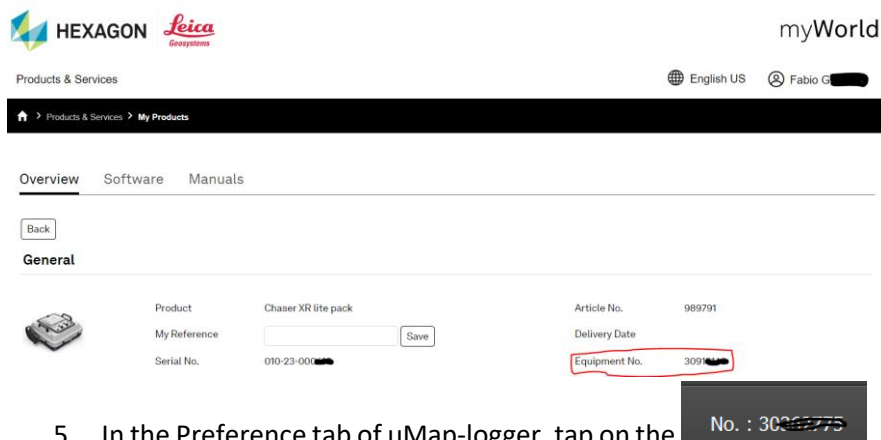
1. The Default Project name **Default Project Name**.
2. The Default Project Folder **Default Projects Folder**.
3. The Measurements System (Metric or Imperial) **Measurement System** (Metric System).
4. The background Theme (Dark or Light) of the uMap-logger Graphical User Interface **Theme** (Dark).
5. The Translation Language (Default is English) **Translations**.
6. The Real Time visualization of the Tomography may be enable/disable **Tomography Processing Enabled during data collection**.

7. The **Air Waves Removal (AWR)** Filter can be enabled to optimize the B Scan view by reducing the interference of external anomalies (Air Waves) . This option can be activated only on Chaser XR dataset.

8. Enabling the **Auto Detect System** function. This function allows the SW to automatically recognizes the HW being connected.

➤ By selecting the **Services** tab, the Remote Diagnostics (**Re.D.**) service may be enabled, through the following steps:

1. Login your **myWorld** page (at this link, the full guide on how to register a new account <https://idsgeoradar.com/customer-area/my-world-customer-portal> and use **myWorld** portal),
2. Enter the **myProducts** area,
3. Select your **Stream DP, Chaser XR or DS4000**, that you have registered earlier,
4. Copy the **Equipment Number** that you find associated with the system (red square in the image below),

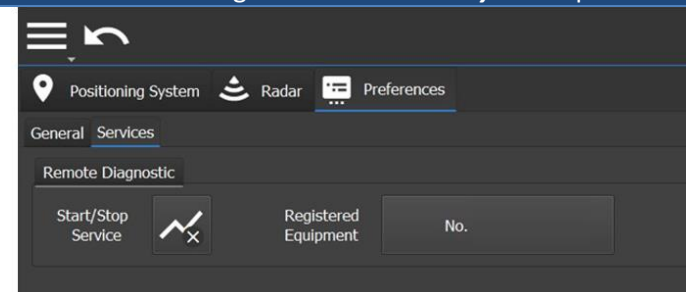


5. In the Preference tab of uMap-logger, tap on the tab that it is under the Service Tab (image below),

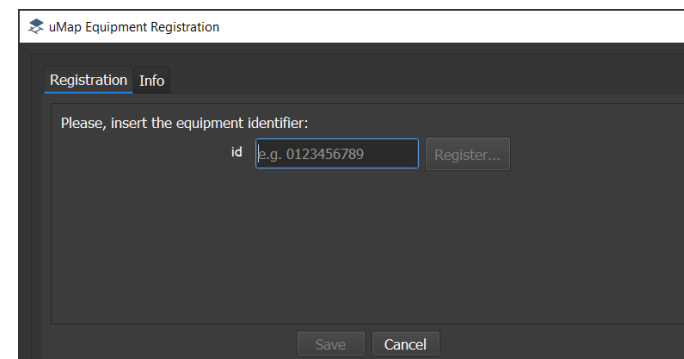
NOTE: All IDS Products are identified by 3 codes:

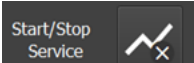
1. The **Product Number** or **Article number** (generally six digits. E.g.: Stream DP Product Number is 970969 and Chaser XR Product Number is 989791)
2. The **Serial Number** specific for one single GPR (example 010-23-123456)
3. The **Equipment Number** (that is a unique combination of the Product Number and Serial Number)

These details are normally found in the documentation (invoice) received once purchased the system. They are used to register Products on **myWorld** portal.



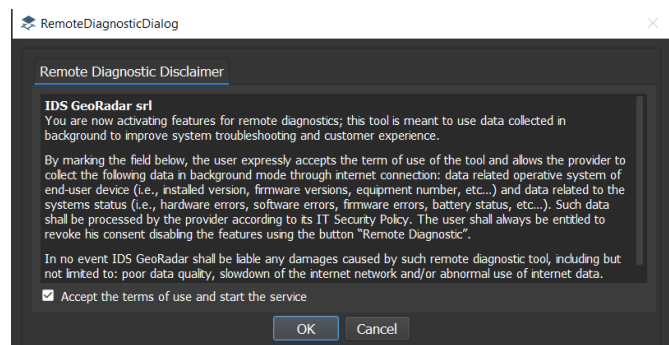
6. Enter the **Equipment Number** in the u Map-Logger Equipment Registration window that shall pop up, and press “**Register**”, then “**Save**”.



1. Press the  button to enable/disable the service. At the activation, the **Remote Diagnostic Disclaimer** window below, shall pop up.



NOTE: Make sure that, before selecting the Equipment ID to register for the Re.D. Service, the correct system has been selected by the drop-down menu of the Radar Configuration tab



2. Select **“Accept”** and then press **OK** to proceed with the Service. Details on the **Re.D.** are described in detail at 6.2.8:

6.2.5 How to update SW and FW

One software (uMap-Logger) and one firmware are involved during a systems SW update.

Whilst the uMap-logger update can be done on the desktop as any other SW update procedure, to update the system FW, User shall:

1. click on  button, from top left of the uMap-Logger window and select the “Update”  command (Fig. 35),

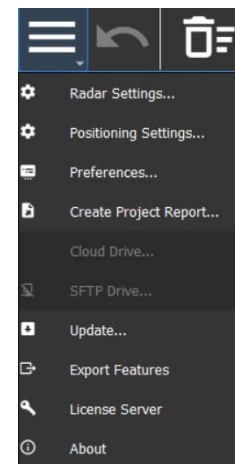


Fig. 35 Hamburger menu Window

2. A new window shall pop up (Fig. 36).

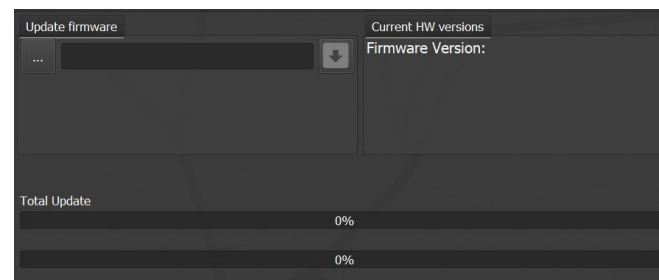


Fig. 36 Update Window

2. From the **Update Firmware** tab: press on  button and select the current version in *.bin format. Once loaded, the update starts by clicking on  button.
3. The version of the firmware can be verified on the top right of the window (Fig. 36).

6.2.6 How to Start a New Project

As the **New Project** icon of Fig. 10 is pressed, User is prompted to the panel of Fig. 37.

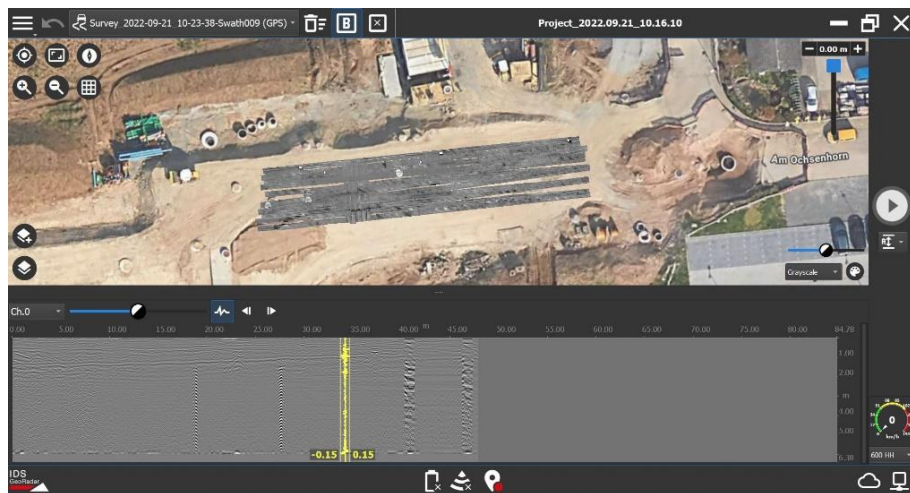


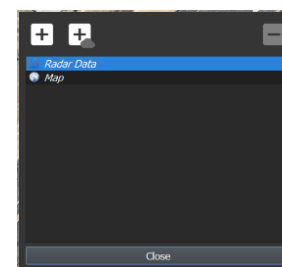
Fig. 37 uMap-Logger New Project Window

The New Project window includes tools, buttons and features, allowing to manage field data acquisitions. These are:

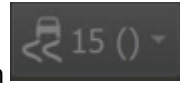
1. **The yellow square**  that shows the System's current position, in real time.
2. **The coverage** and footprint are indicated by the thick blue lines, behind the yellow square . This is updated in real time during the survey acquisition; the light blue dots, inside the thick blue lines, are the (positioning) GNSS, TPS or encoder trigger points.
3. Data acquired are displayed in real-time below the navigation map when the B-Scan is activated by the  icon (Fig. 37).

NOTE: Every time a New Project is initiated, the configuration should be updated by the Radar Status menu icon  and by pressing the  button. This is to make sure the HW deployed, corresponds to the U Map-logger configuration being selected.

4. During data acquisition, the User can set the **orientation** of the view, with respect to the North direction, by pressing the icon .
5. The tool  allows to **Add a Vector Layer** as a base map. The User can add, as background, different vector layers format, such as .kml, .dxf, .gml, .shp, .geojson, .json, , either stored locally or on the cloud, to be selected from the pop-up window below, and by pressing  or , respectively.



6. By selecting the icon , the User can **Add Web Map Service layer**, such as Google Hybrid, Google Maps, Google Satellites, or OpenStreetMap.
7. The tool  allows to actualize the zoom level to the GPR systems current position.

8. Tap  to show grid lines.
9. To Display and centre the entire radar data acquired with respect to centre of the screen, the user shall press  icon.
10. To **Zoom IN** and **Zoom OUT** the icon  shall be used.
11. To Start and stop the acquisition, respectively, the user shall use the button  and . In particular: When the icon  is active the Stream DP or Chaser XR acquisition is running (press this button to stop the acquisition). When the icon  is active the systems acquisition is stopped (press this button to start the acquisition). Users can switch push/pull direction from this icon . Also, the automatic selection of the acquisition direction may be selected.
12. This icon  indicates the speed of the system (odometer) in real time.
13. To **Select** a specific swath, use the drop-down icon .
14. To **Delete** a swath, press the icon .
15. To **hide/show** B-scan can be used the icon .

16. The  icon, allows to display the A-Scan on the B-Scan (Fig. 38) at the single point specified by the left mouse button.

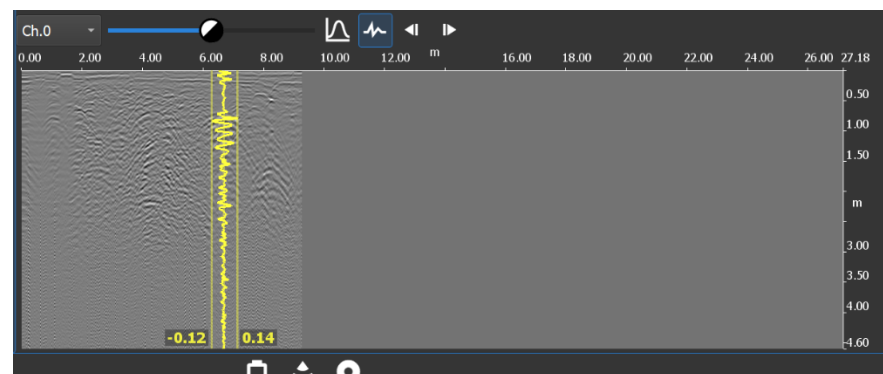
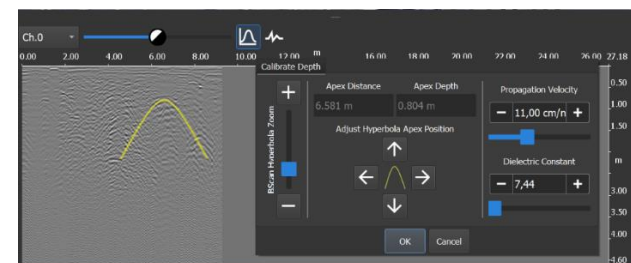


Fig. 38 B-Scan View with A-Scan view (yellow)

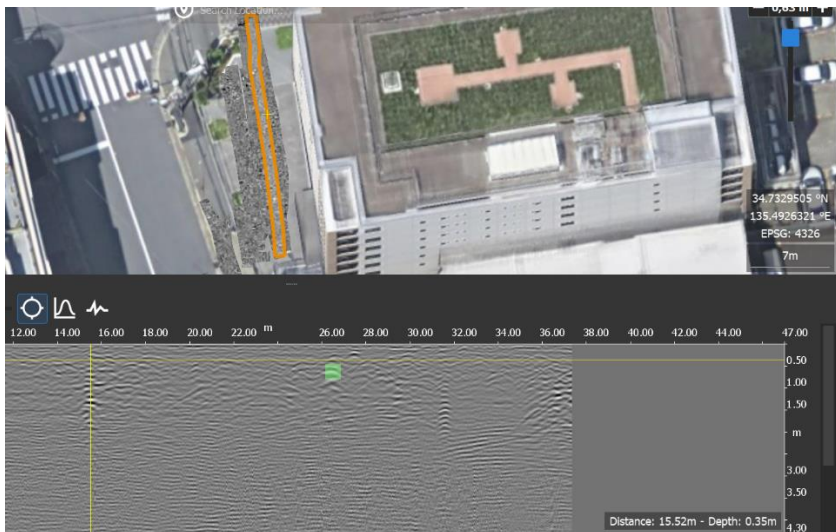
The A-Scan is displayed within two vertical, yellow, parallel lines, ranging between two maximum values of -0.5 to +0.5. The single A-scan to be displayed may be selected by clicking with the left mouse button on the B-Scan, or by clicking the arrow keys  .

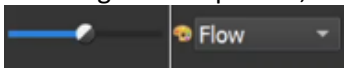
17. The icon  allows the User to adjust the propagation velocity by hyperbola fitting. Icons  and  allows to adjust zoom level of the B-Scan, the Propagation Velocity, and the Dielectric constant. Whereas the symbol  allows to move the hyperbola apex.



18. Fig. 39 B-Scan View with hyperbola fitting

18. The  icon enable the visibility of the cursor on B Scan that is synchronized with C Scan. As cursor is moved throughout the B Scan, information on depth, distance, Northing and Easting, are updated on the right-side panel.



19. To adjust tomography contrast and change colour palette, move the bar and scroll the drop-down menu, .

20. uMap-Logger shows the tomography when the system stops. It is



possible to scroll down the depth by using the slider.

21. Finally, to **Close** the current project the icon  must be selected.

By default, the project ID is set by the format yyyy.mm.dd_hh.mm.ss progressive number. **Project_2020.02.11_16.05.25**

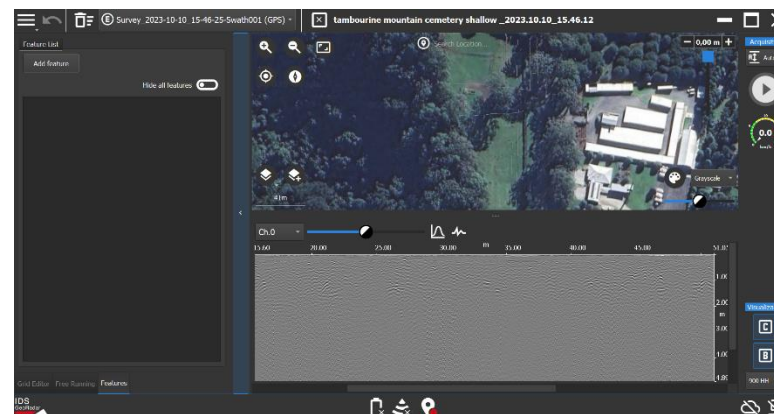
6.2.7 How to insert targets

Targets can be added on B scan and C scan as well while collecting data on site.

To do this:

1. Stop the current Acquisition,

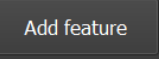
2. Press the “arrow” icon  at the left side of the Graphic Users Interface to enlarge the left panel,

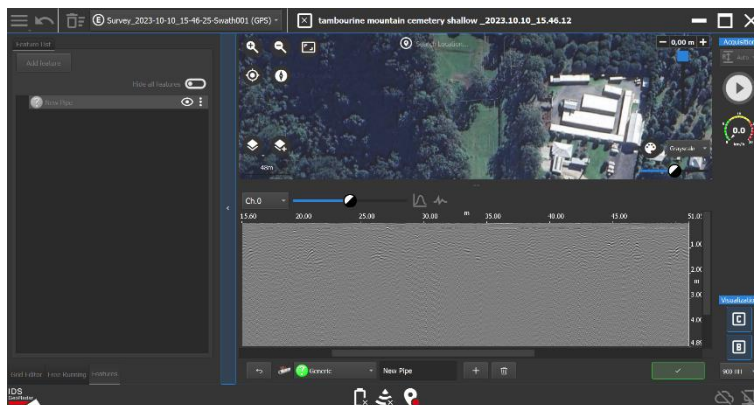


3. Three tabs are present in the left panel (lower side):

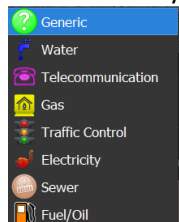
- a. Grid editor
- b. Free Running (not active for DS4000)
- c. Features

4. Select the “Features” tab

5. Press on the  icon,

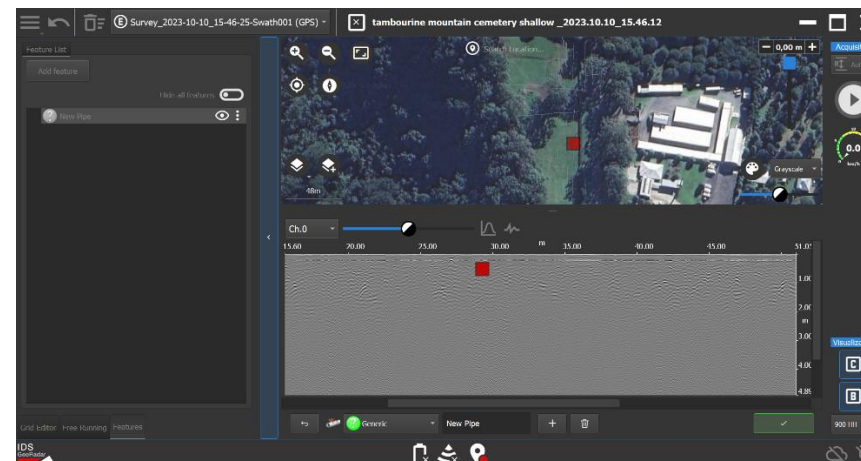


6. Select the type of Feature to add from the drop-down list



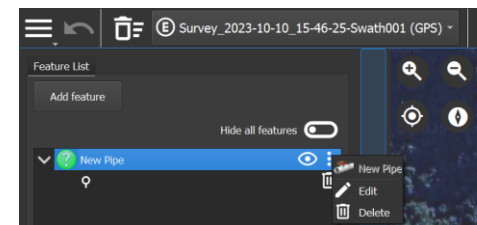
at the lower side of the panel, and press  to add,

7. Select the point of the B Scan or the C scan where to add the feature and press  to save the Feature just add in the Feature list panel on the left.



8. Features present in the List, can be edited, removed or hidden by the

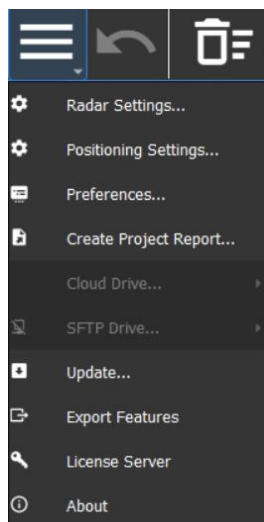
menu below 



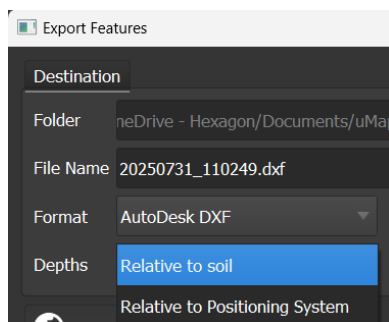
6.2.8 How to export targets

Targets being inserted in the B Scan, can be exported in *.dxf, *.shp, *.kml, IDSGeopmap format, through the steps below:

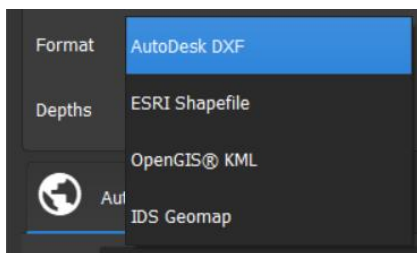
1. Press on the  icon,
2. Select "Export Features",



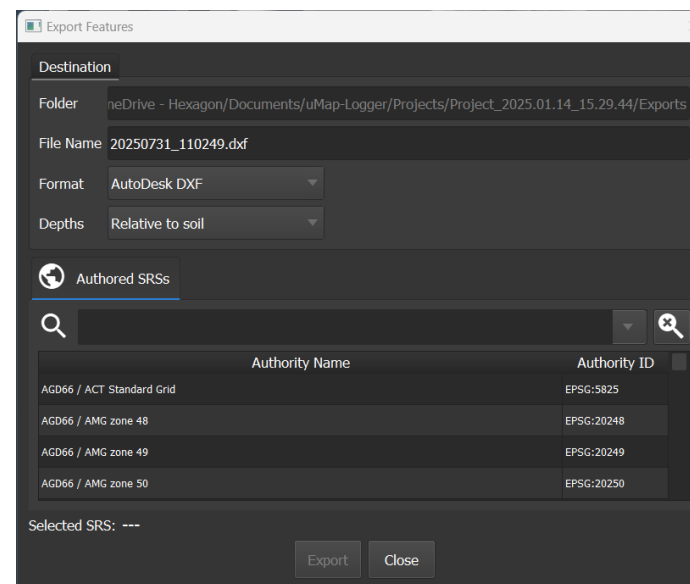
3. Select if features should be export in “Depth” (*Relative to Soil*) or “altitude” with respect to positioning system (*Relative to Positioning System*),



4. Select the format type for the target export (choose between .dxf, Shapefile, .kml, IDS Geomap),



5. Select the reference system,

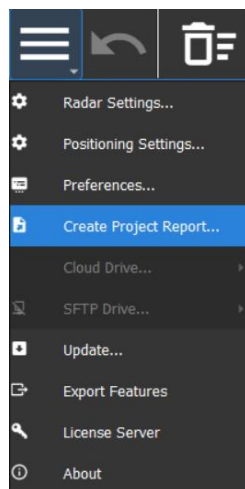


6. Press Export.

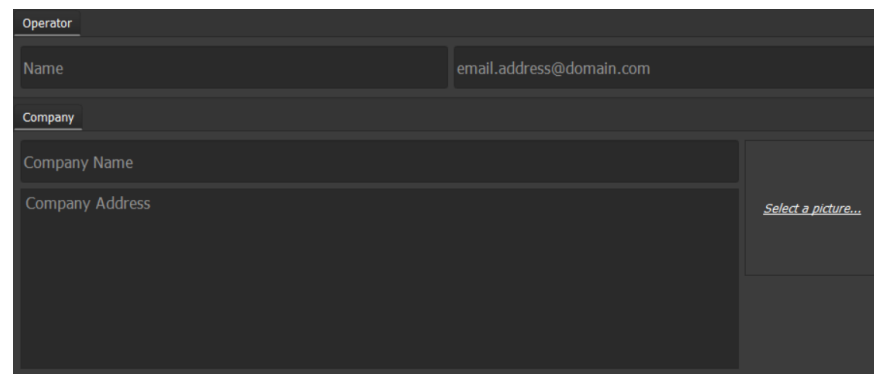
6.2.9 How to Create a Project Report

To Create a Project Report, follow the steps below:

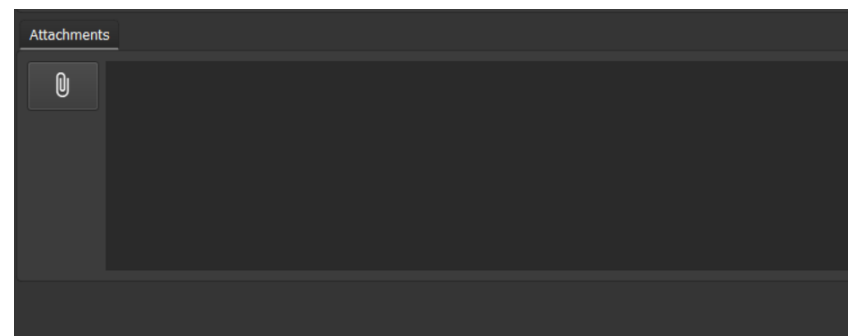
1. Select and open a Project of the Project List by pressing the  icon,
2. Press the  icon on the top left corner,
3. Select the **“Create Project Report”** option from the list below,



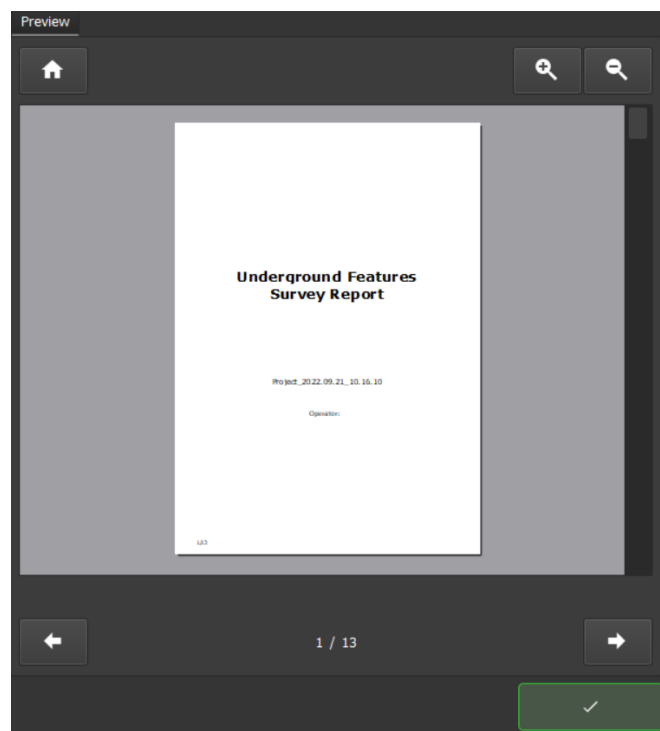
4. Fill out the form with the *Operator* and *Company* information that shall be shown in the final report (*Name*, *e-mail address*, *Company Name*, *Company Address*),
5. Add an image (for example the Company Logo) to the report template, by pressing **“Select a picture”** from the right panel and browsing through the folder where the image is stored,

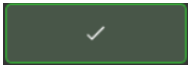

 A screenshot of the 'Create Project Report' form. It has two main sections: 'Operator' and 'Company'. The 'Operator' section has a 'Name' field and an 'email.address@domain.com' field. The 'Company' section has a 'Company Name' field, a 'Company Address' field, and a 'Select a picture...' button on the right.

6. Include attachments in .png, .jpg, .bmp, .tiff, .tif, .gif format, by pressing the  icon,



7. Verify the Report content by the *Preview* window, using the  icon to browse pages, the icon   to zoom in and out, the icon  to restore original zoom level.



8. Press the  icon to choose the folder where to save the survey report in *.pdf format

6.2.10 How to use Remote Diagnostics (Re.D.)

The Basic version of uMap-logger includes the possibility to use the System Remote Diagnostics (**Re.D.**) service, after activating it as illustrated in 6.2.4. Once active, **Re.D.** allows the User to send information on the IDS GeoRadar GPR system deployed onsite (Stream DP or Chaser XR), in real time, to the Cloud that is available for Diagnostics by IDS Technical Support.



NOTE: For the Re.D. to communicate and send data to the Cloud in Real Time, an active internet connection must be always available. If not, GPR systems diagnostic data shall be stored locally on the PC.

Precautions:

Please, make sure internet is available at any time if Re.D. is meant to be used.

Information sent onto the Cloud via **Re.D.**, includes:

- **SESSION**
 - Including: *System Name, Radar Configuration Name, License ID, SW Version.*
- **CONFIGURATION**
 - Including: *Systems Name, Radar Configuration Name, Radar Version, Positioning Settings, Time Sync, Radar Version, Time Sync Settings.*
- **STATUS**
 - Including: *Antenna Status Data, Positioning Status Data, Time Sync Status Data, Flags Status Data, Failure Data, Battery Data.*

The above would allow for the implementation of the following workflow:

1. Device Registration and Service activation,
2. Sending, in Real Time, of GPR Diagnostics data,
3. Remote Diagnostics and Support.

6.2.11 How to work with uMap-Logger Advanced (Optional)

The Advanced version of uMap allows to use additional features to share data between the field crew and the office via commercial Cloud or via File Transfer Protocol. These are:

1. **CDS** (Cloud Data Sharing), (via Google Drive™ or One Drive™), and
2. **SFTP** (Secure File Transfer Protocol).

6.2.11.1 uMap-Logger Advanced License Activation

The default license type of uMap-Logger is the so-called “Basic”. To upgrade to the **Advanced** mode, the User must activate a valid license by using the Leica Client License Manager (CLM) software; to do this please, refer to **Leica Geosystems Software Licensing Introduction & Installation User Manual**.



Note: Before starting the License Upgrade Procedure, make sure that the computer is connected to internet.

The user must open the license Manager software (CLM) then press the ADD Icon (Fig. 40).

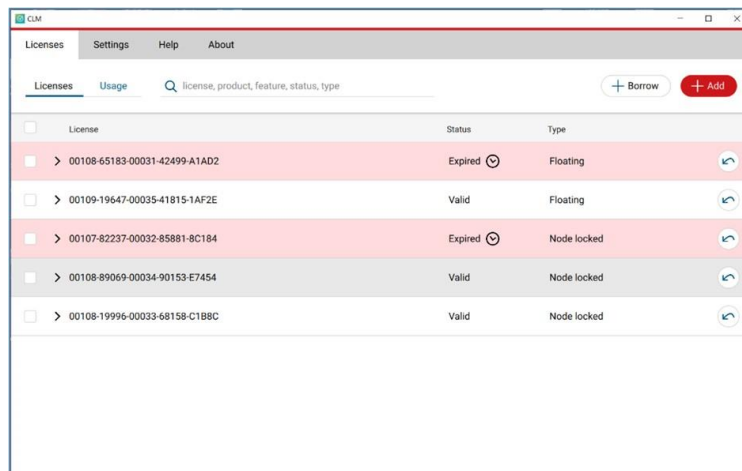


Fig. 40 Client License Manager main page

Then the 25 digits Entitlement ID must be entered in the next screen (Fig. 41), including the dashes between each 5-digit block. Then click “ADD”, again.

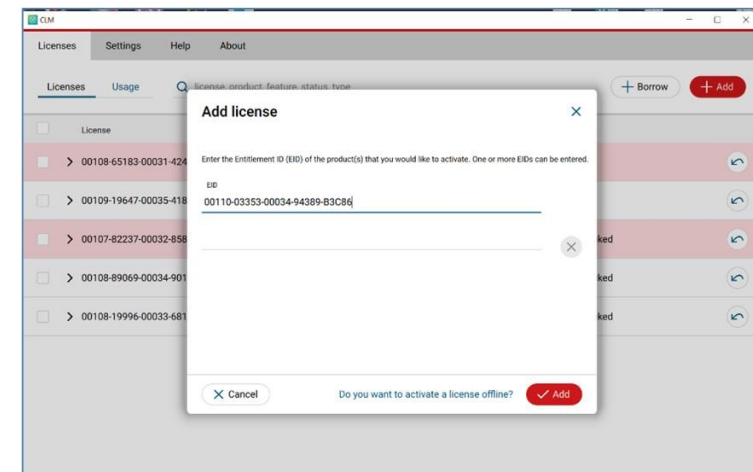


Fig. 41 Adding EID on CLM

Wait for the license activation (Fig. 42) and finally double check if the Entitlement ID is the correct one (Fig. 43).

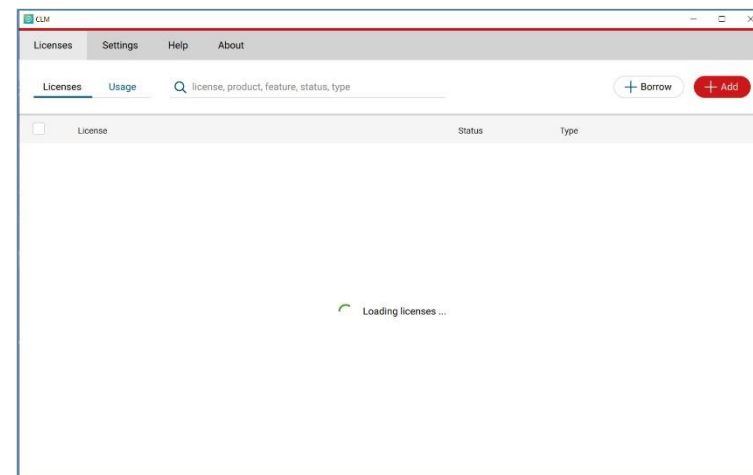


Fig. 42 Waiting for license activation

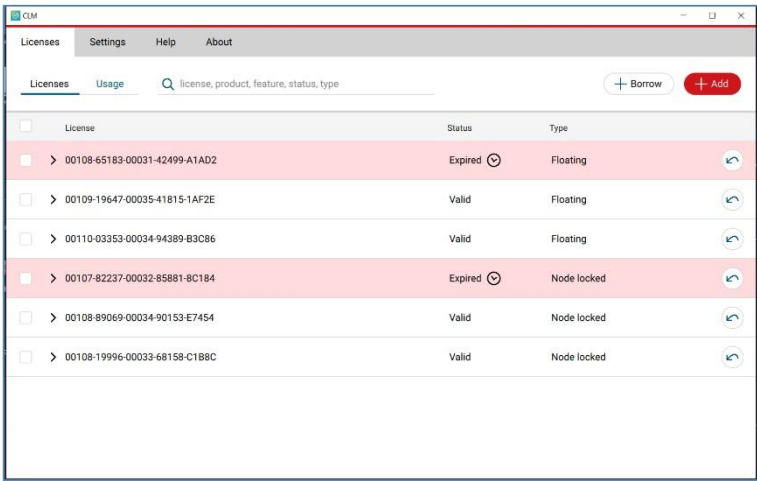


Fig. 43 License activated panel

After the Advanced license activation through an Entitlement ID, the “About” menu will change as in Fig. 44.

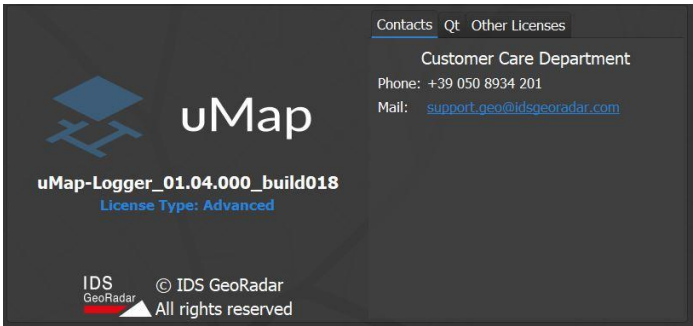


Fig. 44 "About" window of the Help menu after License activation

The About menu, besides the IDS GeoRadar “**Contacts**” tab, contains two more tabs (namely, “**Qt**”, and “**Other licenses**”), where all information on the **Qt libraries** (Fig. 45) and the **Other licenses** (Fig. 46) used to write the SW code, were used.

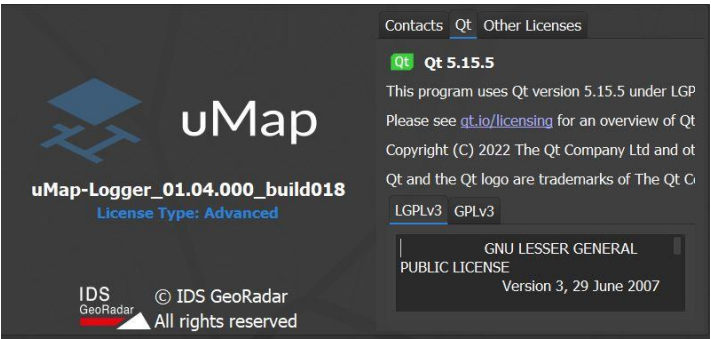


Fig. 45 Qt library tab

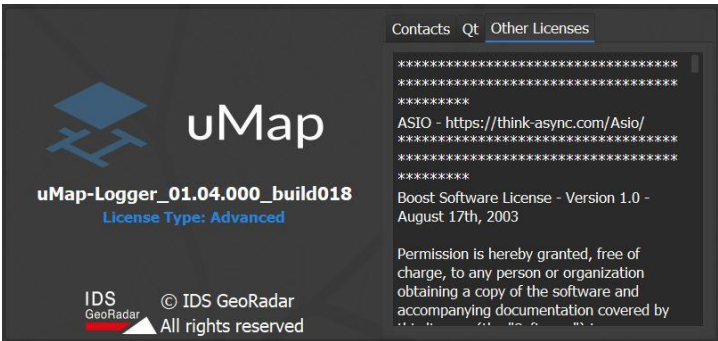


Fig. 46 Other Licenses tab

To connect the uMap-Logger advanced license, the User should specify where this is stored, by tapping the “License Server” key down the Hamburger menu on the top left corner (Fig. 47), and the window (Fig. 48) shall popup.

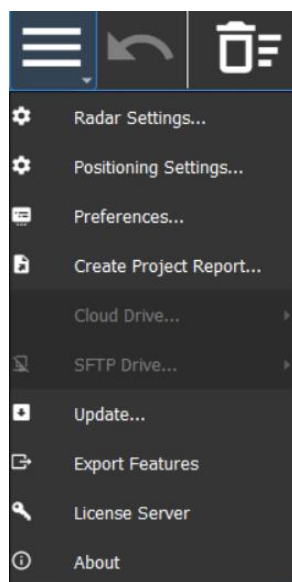


Fig. 47 License server selection

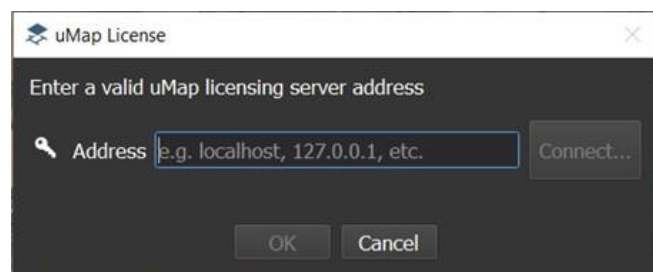


Fig. 48 License server storage

As the correct location is entered (type “localhost” if the license is stored onto the same PC, or the specific IP address if this is installed on a server), press “Connect” and the message shown in Fig. 49, shall popup.

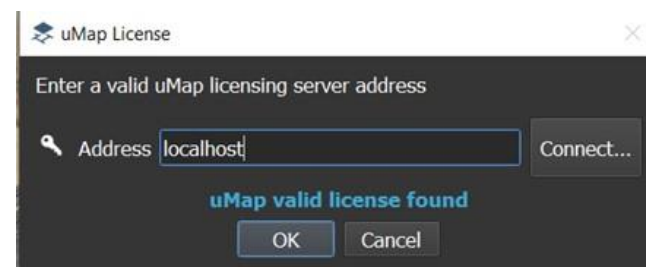


Fig. 49 License connect and validation

6.2.11.2 uMap-Logger Advanced additional features

6.2.11.2.1 Cloud Data Sharing (CDS) (Available for Google Drive™ and One Drive™ account holders only)

Cloud Data Sharing (CDS) is a functionality relying on third-party cloud service, allowing to share data between the field PC connected to Stream DP and Chaser XR systems and the office PC (and vice versa), via commercial Cloud paired with uMap-Logger Advanced. This functionality is meant to enhance the collaboration between field and office, optimizing the whole workflow.

The general idea behind CDS is that the field crew can download files from the third-party cloud service provider onto the field PC, (i.e.: *.dxf, *.kml, *.shp) including reference positioning and geographical information about the working site, or folders (i.e.: survey folder/s) to continue working on a project started earlier and not completed, yet. The office Team, instead, may download a survey uploaded from field and start post-processing far away from field. This workflow is shown in the simplified sketch of Fig. 50.



Fig. 50 Cloud Data Sharing (CDS) Scheme

The commercial cloud that can be linked and used with uMap-Logger Advanced, are Google Drive™ (<https://accounts.google.com/>), and OneDrive™ (<https://www.microsoft.com/en-gb/microsoft-365/onedrive/online-cloud-storage>)

IDS GeoRadar shall not be liable for any damages related/caused by such solution, including but not limited to data breaches and/or losses whatsoever.

If the user does not have any **Google Drive** or **One Drive** account, the first step is to have an active account (for more information, please refer to <https://www.google.com/account/about/> or <https://www.microsoft.com>) and then a **Google Drive** account (<https://drive.google.com/>) or a **One Drive** account, to be paired with uMap-Logger.

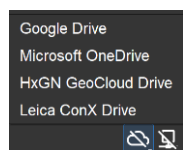
Login the Cloud Drive can be done directly on the Cloud Drive Service Provider or through uMap-Logger, by the following steps, below:

1. On the uMap-Logger main page the Users should click on Cloud icon



(or Open a Project through **Google Drive** or **One Drive**),

2. The User shall be re-direct to the default internet browser to the third-party service provider selected through the icon



3. Then, follow the steps indicated by **the cloud provider** to finalize the procedure.
4. Eventually, the message below shall appear on the browser page,

"Authentication successful. Please return to the Leica field software"

indicating that the process has been successful.



Note

Google Drive™ and One Drive™ are SaaS solution provided by independent third parties, not provided by IDS GeoRadar, and subject to their own relevant Terms and Conditions of Service.

Once the Cloud Drive account is active, and the Cloud Drive is paired with uMap-Logger through the above steps, Field crew and Office Team can share data by pointing to the same Cloud drive having had access with the same credentials.

Login and logout the cloud drive can also be managed by the hamburger menu on the top left corner of the uMap main window, by selecting the "cloud" tab (Fig. 51).

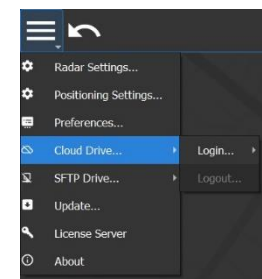


Fig. 51 CDS Login and Logout

CDS by the Office

Although uMap-Logger is developed and optimized as a field data acquisition software rather than a standalone tool, there are functions that may be used by the office. To do this, a uMap-logger stand-alone must be installed onto the office PC.

The upload or download of data can be done by using the standard dashboard features and tools provided by the Cloud drive (Fig. 52).



Fig. 52 google drive dashboard

Or, alternatively, a *stand-alone* version of uMap-Logger Advanced installed in the Office PC can be used to both upload and download files and data folders.

In uploading, folders that may be useful to share with the field crew are Stream DP or Chaser XR survey folder, to be re-opened on field, to continue working.

To do this, select the survey folder to be uploaded, under the open Project

menu (left panel of Fig. 53) and click on the icon  on top of the Projects list.

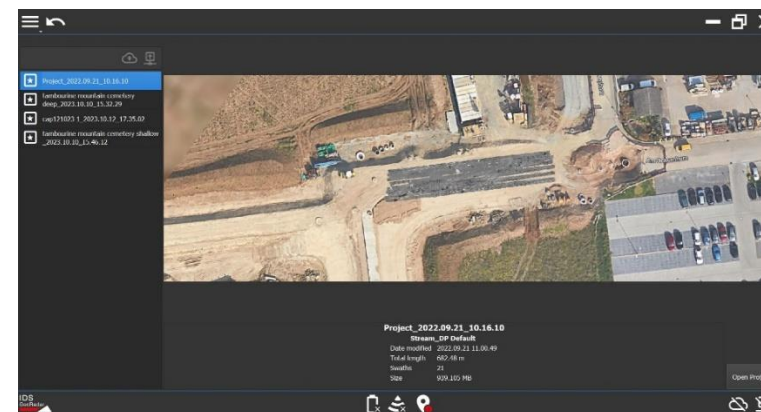


Fig. 53 data upload for CDS

CDS by the Field

Field crew may download and use *.dxf, *.kml, *.shp files uploaded by the office after pairing the uMap-Logger to the Cloud Drive. To do this:

1. Open Project, or start a New Project,
2. Select the Add Vector Layer icon  and then the  icon as illustrated in par. 6.2.6,
3. The cloud drive folder containing shared data shall pop up, and the user can select the files to be downloaded and used on field.

Also, by pressing the Open Project icon (Fig. 54), the window shown in (Fig. 55) shall pop up. This window shall allow the User to upload a project folder on the cloud by selecting the specific project to upload and pressing the  icon.

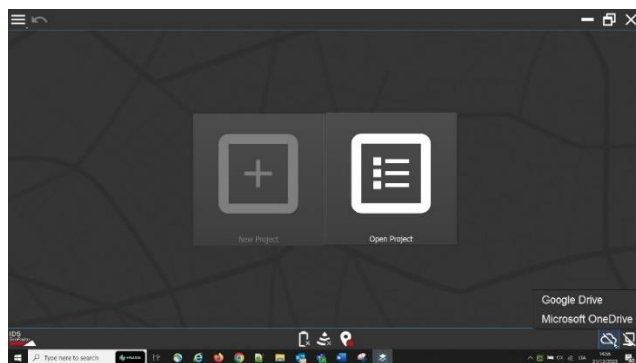


Fig. 54 Open Project

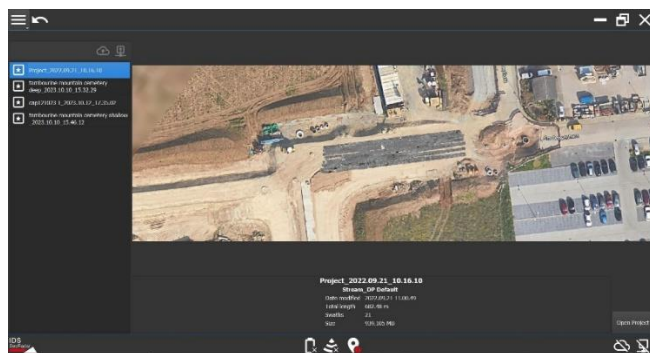


Fig. 55 Cloud drive upload

6.2.11.2.2 Secure File Transfer Protocol (SFTP) Drive

Data collected on site, may also be uploaded onto a remote server via **Secure File Transfer Protocol (SFTP)**. To do this, User should have the following information:

1. Address
2. Port
3. Name
4. Password

To activate the SFTP Drive follow the steps below:

1. Tap on the icon  on the right lower corner of Fig. 55,
2. And the SFTP Authentication window shall pop up

 The screenshot shows the 'SFTP Authentication' dialog box. It has fields for 'Address' (12.23.456.890), 'Port' (123), 'Name' (XXXXXX), and 'Password'. There are 'Login' and 'Cancel' buttons at the bottom.

3. Fill in with the Credentials of the location where User wants to upload the Survey, and press "Login",
4. In the Open Project Window (Fig. 56), select the Project User want to transfer via SFTP and press ,

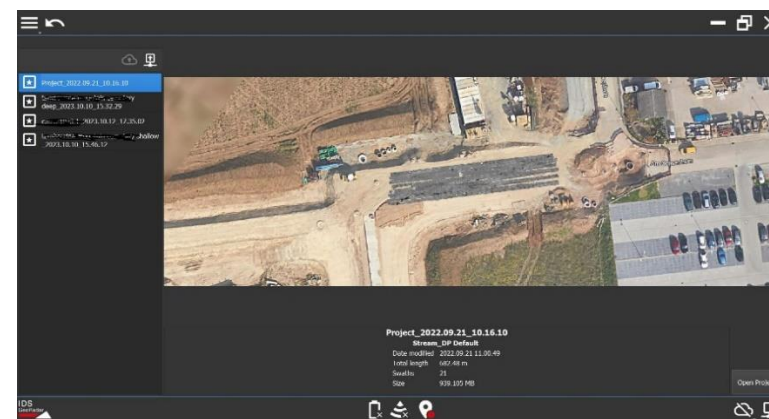


Fig. 56 transfer a Survey via SFTP

5. The SFTP Drive window (Fig. 57), shall pop up,

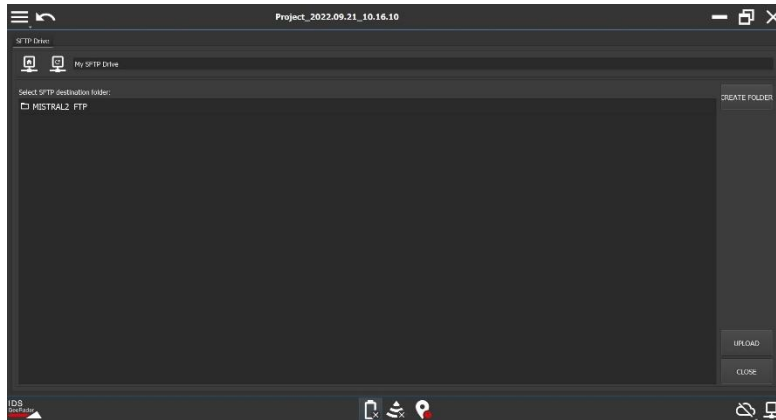
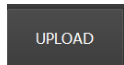
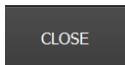


Fig. 57 SFTP Drive window

6. Create a Folder where to upload the Survey, and then press



7. Press



when the Transfer is complete.

7 FREE RUNNING MODE: SIDE NOTES

In GNSS or TPS mode:

1. After pressing the Start button, the B-Scan shall grow regardless of the space run by the Operator and system, as it is triggered by time.
2. After pressing STOP, the B scan shall be correctly resized based on positions given by the positioning system.
3. For computational reasons, the B scan is represented as raw data during acquisition (Fig. 58).
4. After pressing the STOP button, the B-scan is refreshed, resized, and represented as processed data.
5. Markers cannot be added while using GNSS or TPS
6. Examples of possible scenarios when Chaser XR or Stream DP are used in Free Running mode with GNSS or TPS are:
 - a. On small boat for rivers application, or
 - b. On snow, sand, or muddy surface if encoder wheels may not couple correctly to the surface, or
 - c. Contactless applications in general, when the wheel encoder may not trigger correctly the signal, or
 - d. If the wheel encoder chain is, for any reason, damaged.
 - e. Combinations of the above

Line or Grid mode:

1. After pressing the Start button, the B-Scan shall grow regardless of the space run by the Operator and system, as it is triggered by time.
2. After pressing the Start button, the B-Scan shall grow based on speed value set.
3. For computational reasons, the B scan is represented as raw data during acquisition (Fig. 58).

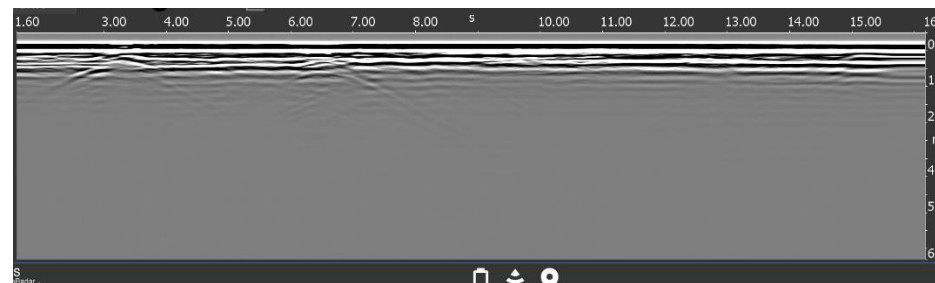
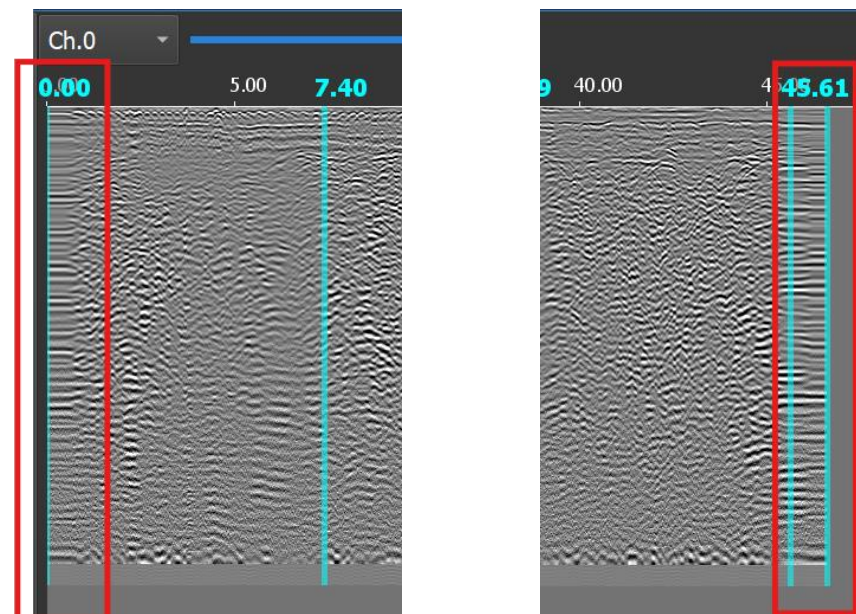


Fig. 58 Raw Data visualization during acquisition

4. After pressing the STOP button, the B-scan is refreshed, resized, and represented as processed data.
5. Because of the lag between the START and STOP action on the screen GUI by the Operator, and the actual START and STOP of the Operator and system, a “stripe” effect can be observed at the beginning and the end of the B scan, after pressing STOP. The longer the lag the wider the “stripe effect” (red rectangle in figure below).



6. Knowing the actual (measured) position of the markers with respect to the start of the line, markers added during acquisition can be edited in post processing and the B scan resized accordingly.
7. The acquisition speed set before the START, influences the progress and visualization of the B Scan.
8. Examples of possible scenarios when Chaser XR or Stream DP are used in Free Running mode in Line mode, are:
 - a. For tunnels applications, or
 - b. On snow, sand, or muddy surface if encoder wheels may not couple correctly to the surface and no GNSS is available, or
 - c. Contactless applications in general, when the wheel encoder may not trigger correctly the signal and no GNSS is available, or
 - d. If the wheel encoder chain is, for any reason, damaged.
 - e. Combinations of the above

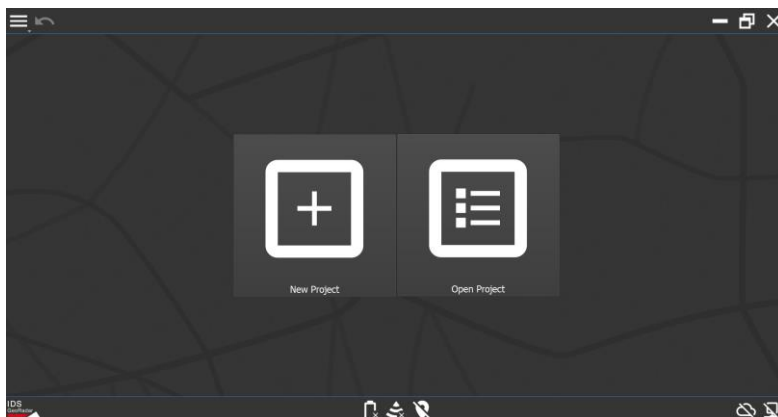


Note: The Free Running mode is NOT available when in use with DS4000

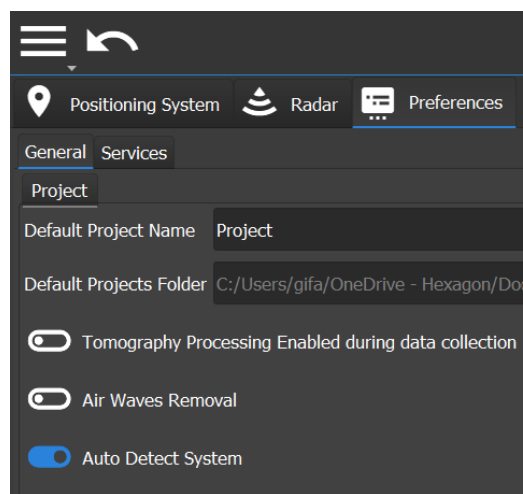
8 UMAP-LOGGER: SUMMARY OF THE WORKFLOW

The complete workflow to run a survey in uMap Logger, includes the following steps:

1. Turn on uMap Logger (Fig. 10),

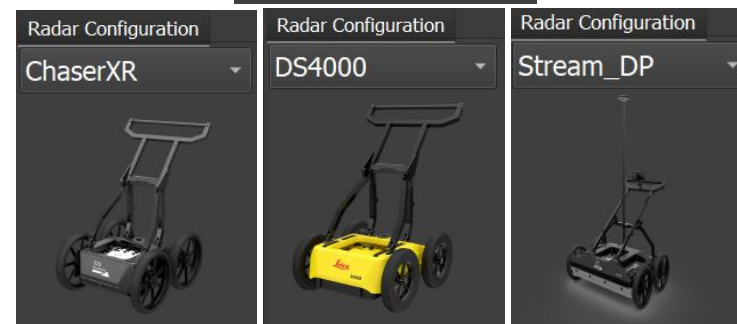
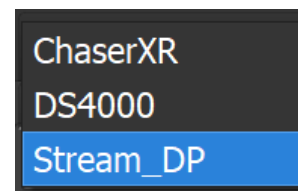


2. Select the **Auto Detect System** function by the **Preferences/General** menu, for the SW to detect automatically the system being used (6.1),

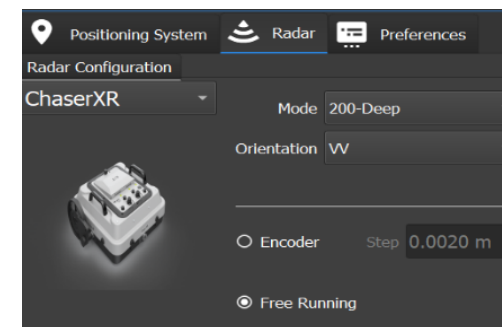


OR

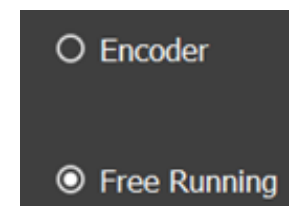
3. Select the System (6.1) from the **Radar Configuration** tab (6.2.1)



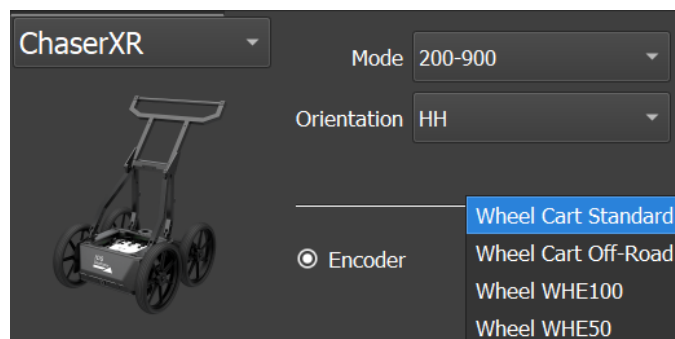
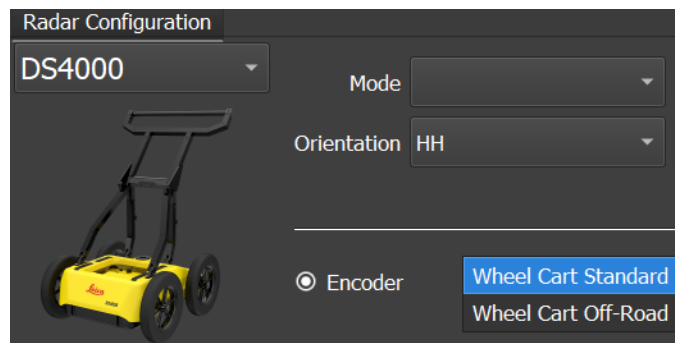
3. Connect and turn on the the GPR System (6.1)
4. Select **Mode** (200-deep or 200-900 Only for Chaser XR) and **Orientation** (HH or VV, for Chaser XR and DS4000) (6.2.1),



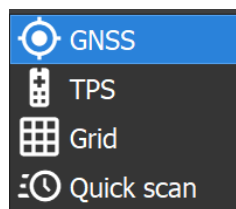
5. Select **Encoder** (with Wheel encoder) or **Free Running** (without Wheel encoder). NOT available for DS4000) (6.2.2),



6. Select the **Encoder type** if working in Encoder mode (only for DS4000 and Chaser XR), depending on wheels being used ,



7. Select Positioning system (6.2.3)



8. Go back to initial page by pressing



9. Start a New Project (6.2.6)



8. Press



to begin.

9. Press



to finish the acquisition.

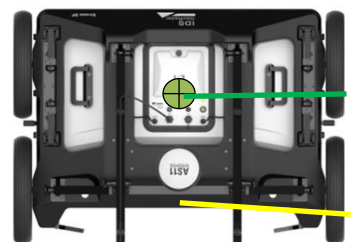
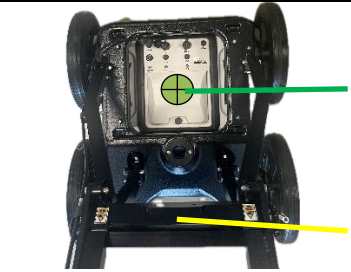
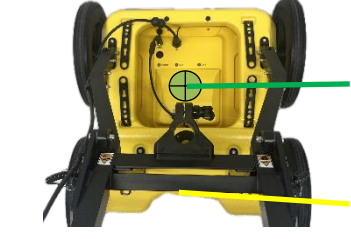
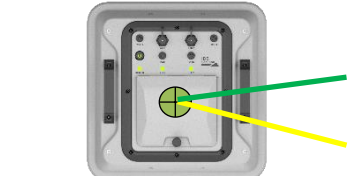
10. Enter targets/Features (6.2.7).

11. Export targets/Features (6.2.8).

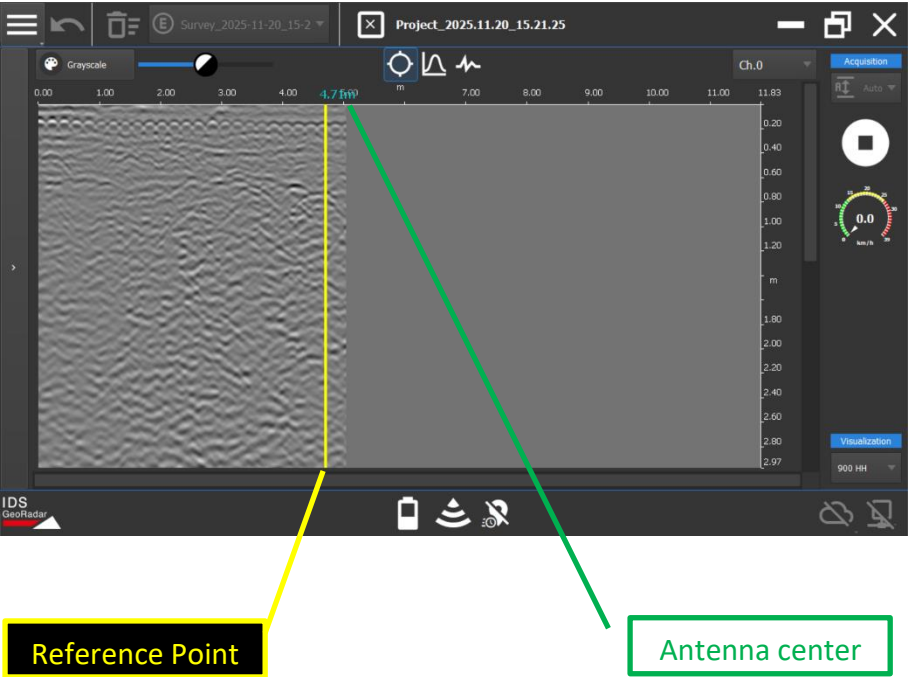
12. Create a Project Report (6.2.9).

9 SYSTEM’S REFERENCE POINT ON B SCAN

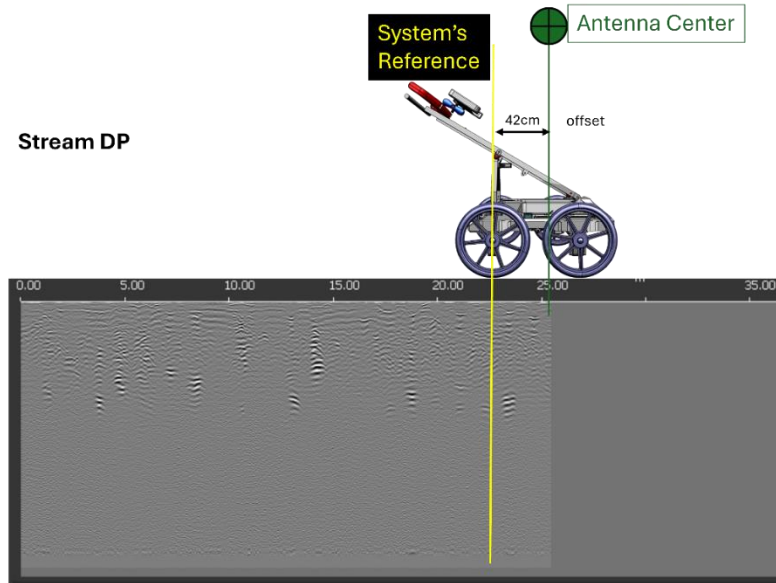
Stream DP, Chaser XR, DS4000, and Opera XR carts, shows a “tick” on their back part, as shown in images below. This, represents the system’s reference point, with respect to the antenna center (the green circle).

Top view	System	Cart Back view
	Stream DP Antenna center Reference Point	
	Chaser XR cart Antenna center Reference Point	
	DS4000/Opera XR Antenna center Reference Point	
	Chaser XR Antenna center Reference Point	

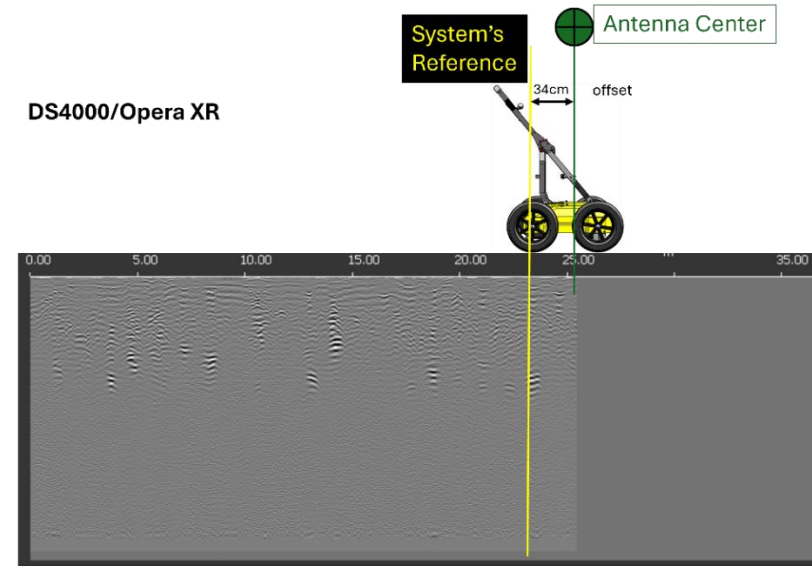
In **uMap Logger**, the **System’s Reference Point** is shown on B Scan during data acquisition, as a vertical yellow line (image below).
Except for Chaser XR with no cart, there is an **offset** between antenna center (where the B Scan starts) and the yellow line in B Scan (The System’s Reference Point).



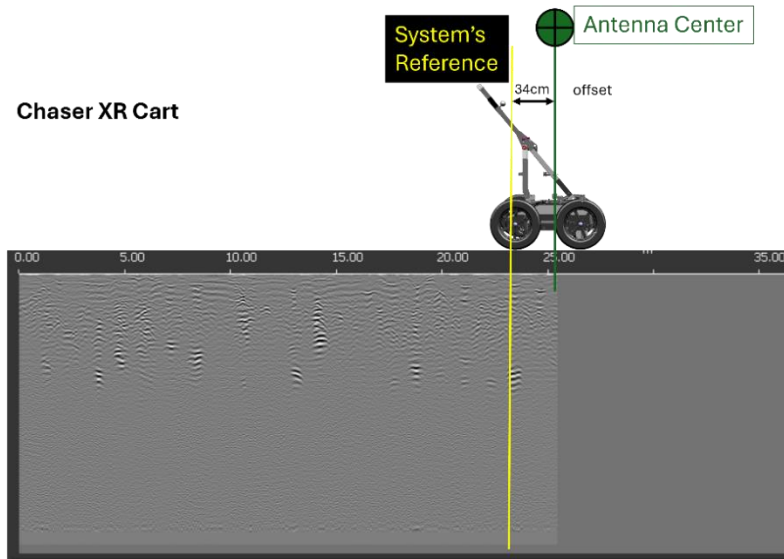
The offset for **Stream DP** is 42cm, and it is shown in image below:



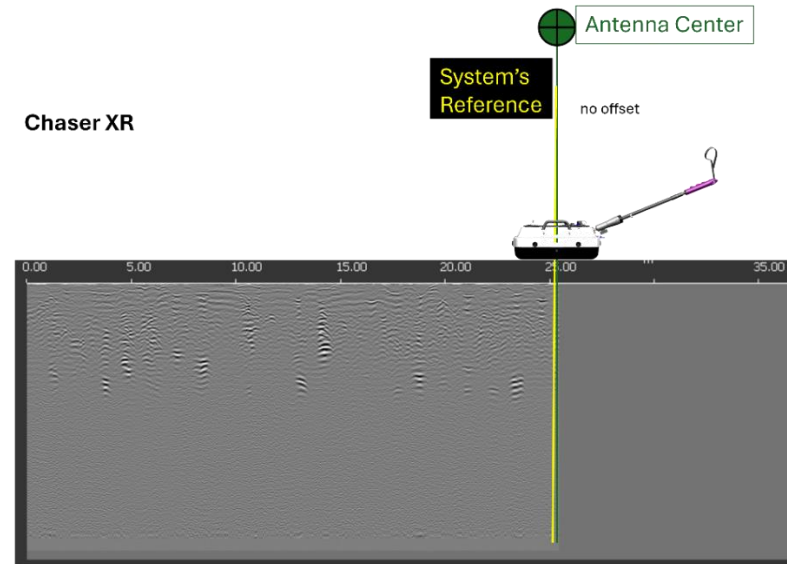
The offset for **DS4000/Opera XR** is 34cm and it is shown in image below:



The offset for **Chaser XR cart** is 34cm and it is shown in image below:



There is no offset for **Chaser XR** with no cart, as shown in image below:



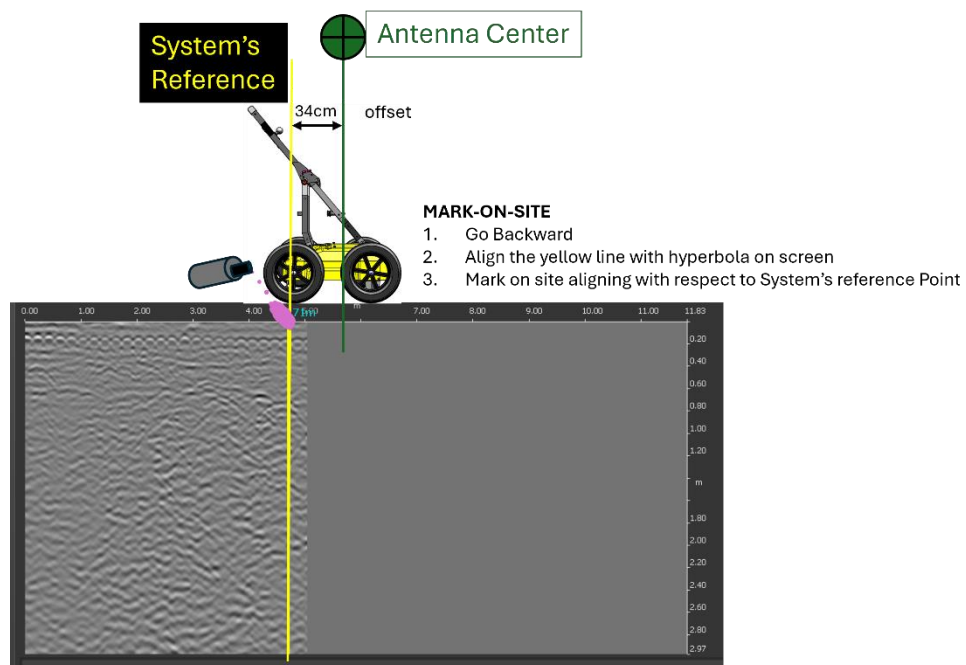
TIP: MARK-ON-SITE

Considering the above, when the GPR Operator wishes to mark on site a feature (Hyperbola) identified on screen, the suggested procedure is:



1. To go backward until the yellow line on screen is on top of the hyperbola of interest,
2. The System's Reference Point placed at the back of the cart (antenna Center in case of Chaser XR with no cart) is now aligned with the feature underneath
3. Mark the ground at the very location corresponding to the Reference Point

This Procedure, is shown in image below.



10 IDS GEORADAR SUPPORT

10.1 myWorld

myWorld platform of Leica Geosystems offers a wide range of services, information, and training material. With direct access to *myWorld*, the User can access all relevant services. The availability of services depends on the instrument model.

SERVICE	DESCRIPTION
myProducts	Add all IDS GeoRadar products that you and your company own and explore your world of Leica Geosystems. View detailed information on your products and update your products with the latest software and keep up to date with the latest documentation.
myService	View the current service status and full-service history of your products in IDS GeoRadar service centres including detailed information on the services performed on your system.
mySupport	Create new support requests for your products that will be answered by your local Support Team. View the complete history of your support requests and view detailed information on each request in case you want to refer to previous support requests.
myLearning	Welcome to the home of Leica Geosystems online learning! There are numerous online courses – available to all customers with products that have valid CCPs (Customer Care Packages).
myTrustedServices	Add your subscriptions and manage users for Leica Geosystems Trusted Services, the secure software services, that assist you to optimise your workflow and increase your efficiency.
mySmartNet	HxGN SmartNet is the GNSS correction service built on the world's largest reference station network, enabling GNSS-capable devices to quickly determine precise positions in the range of one-to-two-centimetre accuracy. The service is provided 24/7 by a highly available infrastructure and professional support team with more than 10 years of experience reliably delivering the service.
myDownloads	Downloads of software, manuals, tools, training material and news for IDS GeoRadar products.

10.2 How to manage return for Service

On occasions, your GPR may need to be returned to factory for repairs. These events, depending on circumstances, may be handled under maintenance or warranty conditions.

IDS GeoRadar, implements an internal ticket system to handle the repair of its Products worldwide. In this respect, for the global IDS Customer Support to be able to troubleshoot in case of need, it is necessary identifying the GPR system first hence, it is mandatory providing information including Serial number and Product Number of the system or part of it that need repair.

This information may be found on a label placed on top of the system (e.g.: inside the battery housing, on top of the antenna, on the side of the rudder, on top of the SCU, etc....). An example of this label is illustrated in Fig. 59.



Fig. 59 Example of IDS GPR system label

In any event, the workflow to follow up on a support request is:

1. Contact, by using the local Support mailing list, your local representative of IDS Customer Support, or enter a support case into the *mySupport* page of *myWorld*, Indicating the following:
 - a. Type of System or part of it (defined by Product Number or Article number, that is PN 859224 in Fig. 59)
 - b. Serial Number (SN 020-23-000543, in Fig. 59)
 - c. Type of issue

2. The local Representative of IDS Customer Support or the Central IDS Customer Support shall follow up on the case by a first level, remote troubleshooting to assess the system and the possible issue.
3. If a further, direct assessment of the system is necessary, the system shall be returned for analysis. Based on this, a repair under warranty or maintenance conditions, may be carried out. Depending on cases and the type of possible repair, return may be towards the local Representative of IDS or to the IDS GeoRadar main factory.
4. Prior return of the goods, wait for the IDS logistic office to provide a reference number for the return case and shipping instructions.
5. As the system reaches IDS repair premises, this is analyzed to find the root cause of the issue, and repaired ONLY after signed approval of the quotation, in case where the warranty does not apply. The status of the repair may be checked onto the *mySupport* page of *myWorld*.
6. At the end of the repair, the system is shipped back to the shipping address indicated at the beginning of the procedure, and this is conducted based on contractual agreements.



NOTE: The above, is meant solely as a broad indication of the workflow for the return of a system for repair.

Precautions:

Please, refer to the Contractual Terms and Conditions for more detailed information.

APPENDIX A DISCLAIMER

1. General

- i. The present Disclaimer applies to all products designed, produced and distributed by IDS GeoRadar S.r.l., its Subsidiaries, Affiliated and authorized Distributors (the “Products”). IDS GeoRadar S.r.l. reserves full ownership and intellectual property rights of any “Information” contained in this Disclaimer including Trade Marks and Graphics. No part of this Disclaimer may be used or reproduced in any forms without the prior written agreement of IDS GeoRadar S.r.l.
- ii. In the event that any provision of this Disclaimer may be invalid, unlawful or incapable of being enforced by a rule of law, all other provisions shall, nonetheless, remain in full force and effect. Failure to either enforce or exercise any right, privilege, or legal remedy at any time, any provision contained in this Disclaimer, shall not be deemed a waiver of such provisions or right, remedy, or privilege.
- iii. This Disclaimer shall be interpreted, governed, construed and enforced in accordance with the laws of Italy. User/Buyer hereby consents to the exclusive jurisdiction of Pisa.

2. Initial Precautions for Setting-up and Use of the Products.

- i. The User/Buyer, for setting-up and using the Products, shall consult the official documentation provided by IDS GeoRadar S.r.l. for the Products (“Reference Documentation”) and carefully ascertain the compliance with national laws and requirements, which may limit or even forbid their use.
- ii. For Products which shall operate by circulation in Public Areas/Roads, with or without moving traffic, Buyer/User shall verify the approval of local authority and/or site’s owner according to their specific procedures. IDS GeoRadar S.r.l. shall not be liable for any direct, indirect, special, incidental or consequential damages or injuries, including without limitation, lost revenues or lost profits,

resulting by un-authorized use of the Products in Public Areas/Roads.

iii. Buyer/User further warrants:

- that these Products are not being used, in the design, development, production or use of chemical, biological, nuclear ballistic weapons. Buyer/ User will defend, indemnify and hold IDS GeoRadar S.r.l. harmless against any liability (including attorney’s fees) for non-compliance with the terms of this article.
- That, if IDS GeoRadar S.r.l. requires that Buyer/User shall carry out a training with reference to some Product categories, no operation or use of the Products shall be started before its designated Operator/s has got the User Certificate, as defined by IDS GeoRadar S.r.l. specific procedure which the Buyer confirms to know and accept.

iv. For Products which include specific “Operational” software with automatic data processing and analysis “Tools”, e.g. the IBIS Products and Hydra Products, User shall be aware that the results provided by these “Tools” may be not error free. User that completely relies on the outcomes provided by these Tools only, does it at his own risk.

v. In no event IDS GeoRadar S.r.l. shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the use of the Products, either separately or in combination with other products or relied upon the results provided by the above “Tools”.

3. Disclaimer for the “Use” of the Products.

- i. The User shall follow the instructions provided by IDS GeoRadar S.r.l. in its official “Reference Documentation” for the Product, in particular the User’s Technical Manual which contains all the specific steps and recommendations for a correct setting-up and use of the Product.

- ii. In no event IDS GeoRadar S.r.l. shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the missed or incomplete observance of the instructions and prescriptions for the use of the Products, either separately or in combination with other products, including but not limited to the following main aspects:
 - a. Use of IDS GeoRadar S.r.l. Products outside its limitation of use, without proper and adequate scientific/technical knowledge or without specific training.
 - b. Use of results/outcomes of the measurements performed by the Product aimed to safety aspects without using adequate control procedures and assessment by skilled personnel.
 - c. Opening of the Equipment (for HW Products) without express written authorization of IDS GeoRadar S.r.l.;
 - d. Unauthorized changes and additions to the Products.
 - e. Use of the Products connected to suspected non-working equipment or with equipment (mainly PC) having characteristics not in compliance with the required specifications of IDS GeoRadar S.r.l. not expressly authorized by IDS GeoRadar S.r.l.;
 - f. Poor or faulty operation of the electrical and telecommunication networks not directly managed by IDS GeoRadar S.r.l. or its delegates.
 - g. Poor or faulty operation Software/Hardware of the third parties connected with IDS GeoRadar S.r.l. Equipment.
 - h. Poor or faulty operation of the Products due to Software Virus which infected the Products after their delivery.
 - i. Use of the Products which have encountered suspected manumissions, accidents, electrostatic shocks, flashes, fire, earthquake, flooding or other natural disasters or unexpected events.
- j. Use or storage of the Products outside the limits of the “Operational Temperature Range” specified by IDS GeoRadar S.r.l.

APPENDIX B CONTACTS AND TECHNICAL SUPPORT

The IDS GeoRadar Technical Support team is available to answer any question about:

- general enquiries
- request of advertising material
- technical issues
- any other business
- As well as listening to your suggestions too.

Please, feel free to contact us sending an e-mail to:

support.geo@idsgeoradar.com

Or call the Number: +39 050 8934 200 - Customer Care Office

Address:

IDS GeoRadar srl

A. Righi 6, 6 A, 8, Località Ospedaletto - 56121 PISA - ITALY

